

ESP32-S31 Series

Datasheet Pre-release v0.5

MCU with one RISC-V 32-bit dual-core high-performance microprocessor and one single-core low-power microprocessor

2.4 GHz Wi-Fi 6, Bluetooth® 5.4 (LE), Bluetooth® Classic, Zigbee, and Thread (802.15.4)

Concurrent access to flash and PSRAM

60 GPIOs

QFN80 (8 × 8 mm) Package

Including:

ESP32-S31NRV16

ESP32-S31NRV32

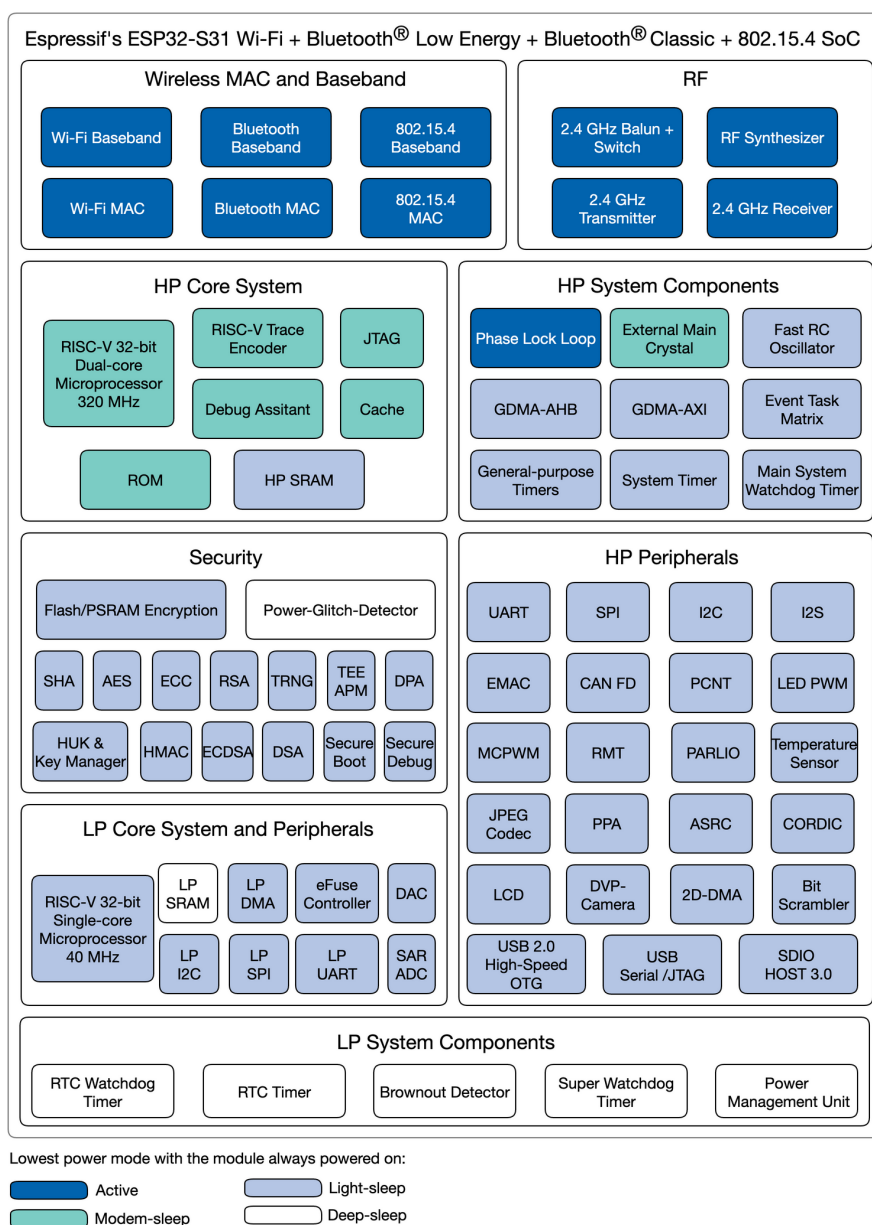


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Product Overview

The ESP32-S31 SoC (System on Chip) supports 2.4 GHz Wi-Fi 6, Bluetooth® 5.4 (LE), Bluetooth® Classic, Zigbee 3.0, and Thread 1.4. The MCU consists of a High Performance (HP) system and a Low Power (LP) system. The HP system contains a RISC-V dual-core CPU and rich peripherals, while the LP system contains a low-power RISC-V single-core CPU and various peripherals optimized for low-power applications. The chip also integrates wireless baseband and MAC (Wi-Fi, Bluetooth, and 802.15.4), an RF module, and numerous peripherals. The chip is designed for applications requiring high-throughput wireless connectivity and higher processing performance.

The functional block diagram of the SoC is shown below.



ESP32-S31 Functional Block Diagram

For more information on power consumption, see Section [4.1.4.6 Power Management Unit](#).

Features

Wi-Fi

- 1T1R in 2.4 GHz band
- Operating frequency: 2412 ~ 2484 MHz
- IEEE 802.11ax-compliant:
 - 20 MHz-only non-AP mode
 - Uplink and downlink OFDMA to enhance connectivity and performance in congested environments for IoT applications
 - Downlink MU-MIMO (multi-user, multiple input, multiple output) to increase network capacity
 - Beamformee that improves signal quality
 - Spatial reuse to maximize parallel transmissions
 - Target wake time (TWT) that optimizes power saving mechanisms
- Fully compatible with IEEE 802.11b/g/n protocol:
 - 20 MHz and 40 MHz bandwidth
 - Data rate up to 150 Mbps
 - Wi-Fi multimedia (WMM)
 - TX/RX A-MPDU, TX/RX A-MSDU
 - Immediate block ACK
 - Fragmentation and defragmentation
 - Transmission opportunity (TXOP)
 - Automatic beacon monitoring (hardware TSF)
 - Four virtual Wi-Fi interfaces
 - Simultaneous support for Infrastructure BSS in Station mode, SoftAP mode, Station + SoftAP mode, and promiscuous mode

Note that when ESP32-S31 scans in Station mode, the SoftAP channel will change along with the Station channel

 - Antenna diversity
 - 802.11mc FTM

Bluetooth

Bluetooth LE

- Full support for the Bluetooth 5.4 (LE) core specification
- Bluetooth Mesh 1.1
- LE Audio (Isochronous Channels, BIS and CIS)

- Direction Finding (AoA/AoD)
- Periodic Advertising with Responses (PAWR)
- LE Connection Subrating
- LE Power Control
- Speed: 125 Kbps, 500 Kbps, 1 Mbps, 2 Mbps
- LE Advertising Extensions and Multiple Advertising Sets
- Simultaneous Operation of Broadcaster, Observer, Central, and Peripheral Devices

Bluetooth Classic

- Data rates: Basic rate 1 Mbps, Enhanced data rate 2 Mbps, 3 Mbps
- Asynchronous connection-oriented (ACL) links, synchronous connection-oriented (SCO) and enhanced synchronous connection-oriented (eSCO) links
- Voice coding formats: A-law, μ -law, CVSD and transparent data
- Channel Classification and adaptive frequency hopping (AFH)
- Traditional power control and enhanced power control
- Secure simple pairing (SSP)
- Data encryption: E0 and AES-CCM
- Secure Connections
- Sniff mode and Sniff Subrating mode
- Role switching
- Active Peripheral Broadcast
- Multiple connections and scatternet
 - Up to 2 synchronous links in the same piconet
 - As a central device, supports up to 7 ACL links in the same piconet
 - Up to 3 piconets simultaneously (1 central device, 2 peripheral devices)
- Maximum output power meets Power Class 1 device requirements
- Hardware-implemented clear channel assessment (CCA)

IEEE 802.15.4

- Compliant with IEEE 802.15.4-2015 protocol
- OQPSK PHY in 2.4 GHz band
- Data rate: 250 Kbps
- Thread 1.4
- Zigbee 3.0

- Matter
- Application-layer protocols (HomeKit, MQTT, etc.)

CPU and Memory

- 32-bit RISC-V dual-core processor up to 320 MHz for the HP system
- 32-bit RISC-V single-core processor for the LP system:
 - Clock speed: up to 40 MHz
 - Two-stage pipeline
- CoreMark® score (dual-core):
 - at 320 MHz: 2195.20 CoreMark; 6.86 CoreMark/MHz
- Single-precision floating-point unit (FPU) per core
- 128-bit data bus with SIMD commands (supported by only one core)
- Sv32 two-level page-table address translation
- Private instruction cache (I-cache) for each core and shared data cache (D-cache)
- 320 KB ROM (accessed through dedicated ROM cache)
- 512 KB HP SRAM
- 32 KB LP SRAM
- External PSRAM Interface: 250 MHz 8-bit DDR PSRAM (in-package only)
- Concurrent access to flash and PSRAM
- Supports multiple SPI interfaces for external flash/RAM:
 - Dedicated interfaces: SPI, Dual SPI, Quad SPI, Octal SPI, QPI, OPI
 - General-purpose interfaces: SPI, Dual SPI, Quad SPI, QPI (supports multiple flash or RAM devices)
- Supports Flash In-Circuit Programming (ICP)

System DMA

- General DMA controller (GDMA):
 - GDMA-AHB: 5 TX channels and 5 RX channels on the HP side; 2 TX channels and 2 RX channels on the LP side
 - GDMA-AXI: 3 TX channels and 3 RX channels

Peripherals

- 60 Programmable GPIOs
 - 4 Strapping pins
 - 6 Pins for off-package flash connection
- Image and Audio Processing:

- JPEG Codec
- Pixel-Processing Accelerator (PPA)
- 2D-DMA controller
- Audio Sample Rate Converter (ASRC)
- CORDIC Accelerator
- LCD and Camera Controller
- Digital Interfaces and Peripherals:
 - 4 UART + 1 low-power UART (LP UART)
 - 6 SPI + 1 low-power SPI (LP SPI)
 - * 2 SPI dedicated to in-package PSRAM
 - * 2 SPI dedicated to off-package flash
 - * 2 general-purpose SPI
 - * 1 low-power SPI
 - 2 I2C + 1 low-power I2C (LP I2C)
 - 2 I2S interfaces with hardware-level Bluetooth Audio
 - 2 pulse counters
 - USB 2.0 High-Speed OTG
 - USB Serial/JTAG controller
 - 1000 Mbps Ethernet MAC
 - CAN FD controller, compatible with ISO 11898-1:2015
 - SDIO host controller with 2 slots
 - 2 LED PWM controllers, up to 8 channels
 - 4 Motor Control PWM (MCPWM)
 - Infrared Remote Control (TX/RX)
 - Parallel IO Controller (PARLIO)
 - BitScrambler
- Analog Signal Processing:
 - Touch Sensor
 - Temperature Sensor
 - 2 x 12-bit SAR ADC, up to 16 channels
 - 2 x DAC
 - Analog Voltage Comparator (1 Reference Voltage Input + 3 Comparison Voltage Inputs)
- Timers:

- 4 x 54-bit General-Purpose Timers
- 1 x 52-bit System Timer
- 3 Watchdog Timers

Power Management

- Power Management Unit (PMU)
- Brown-out Detector

Security

- Cryptographic Hardware Acceleration:
 - AES accelerator
 - ECC accelerator
 - HMAC accelerator
 - RSA accelerator
 - SHA accelerator
 - RSA digital signature peripheral (RSA_DS)
 - ECDSA digital signature peripheral (ECDSA_DS)
- Secure Debug Controller (SDC)
- External Memory Encryption and Decryption (XTS_AES)
- True Random Number Generator (TRNG)
- Key Manager
- Power Glitch Detector
- Secure Boot
- Side-Channel Attack Resistance (DPA)
- Access Permission Management (APM) and Trusted Execution Environment (TEE) Controller

RF Module

- Antenna switches, RF balun, power amplifier, low-noise receive amplifier
- Up to +21 dBm of power for an 802.11b transmission
- Up to +20 dBm of power for an 802.11ax transmission
- Up to -105 dBm of sensitivity for Bluetooth LE receiver (125 Kbps)

Applications

With low power consumption, ESP32-S31 is an ideal choice for IoT devices in the following areas:

- Smart Home
- Industrial Automation
- Health Care
- Consumer Electronics
- Smart Agriculture
- POS Machines
- Service Robot
- Audio Devices
- Generic Low-power IoT Sensor Hubs
- Generic Low-power IoT Data Loggers
- Cameras for Video Streaming
- USB Devices
- Speech Recognition
- Image Recognition
- Wi-Fi + Bluetooth Networking Card
- Touch and Proximity Sensing

Note:

Check the link or the QR code to make sure that you use the latest version of this document:
https://www.espressif.com/documentation/esp32-s31_datasheet_en.pdf



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1 ESP32-S31 Series Information

1.1 Nomenclature

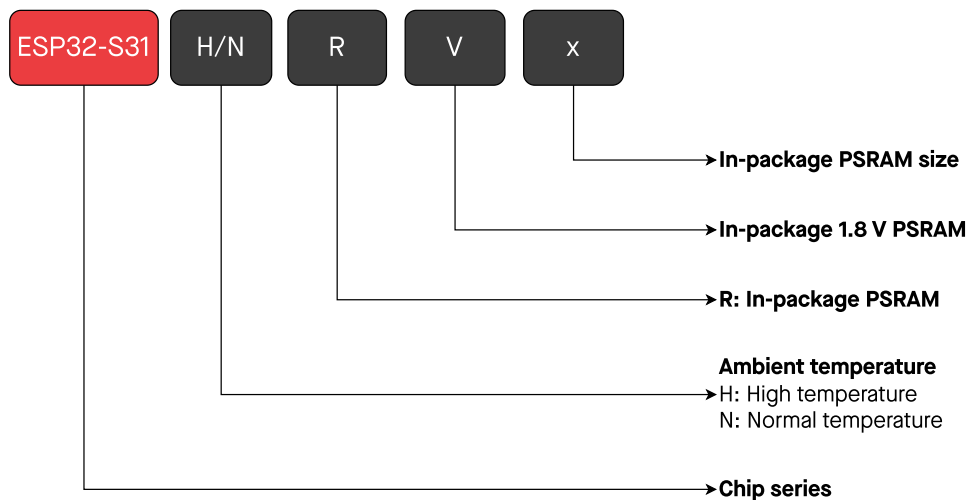


Figure 1-1. ESP32-S31 Series Nomenclature

1.2 Series Information

Table 1-1. ESP32-S31 Series Information

Part Number ¹	Ambient Temp. ² (°C)	In-Package PSRAM	Package
ESP32-S31NRV16	–40~85	16 MB (Octal SPI) ³	QFN80 (8×8 mm)
ESP32-S31NRV32	–40~85	32 MB (Octal SPI)	QFN80 (8×8 mm)

¹ For details on chip marking and packing, see Section [7 Packaging](#).

² Ambient temperature specifies the recommended temperature range of the environment outside an Espressif chip.

³ For details about SPI modes, see Section [2.6 Pin Mapping Between Chip and Flash](#).

2.2 Pin Overview

The ESP32-S31 chip integrates multiple peripherals that require communication with the outside world. To keep the chip package size reasonably small, the number of available pins has to be limited. So the only way to route all the incoming and outgoing signals is through pin multiplexing. Pin muxing is controlled via software programmable registers.

The ESP32-S31 chip has the following types of pins:

- **IO pins** with the following predefined sets of functions to choose from:
 - Each IO pin has predefined **IO MUX functions** – see Table 2-3 *IO MUX Functions*
 - Some IO pins have predefined **LP IO MUX functions** – see Table 2-5 *LP IO MUX Functions*
 - Some IO pins have predefined **analog functions** – see Table 2-7 *Analog Functions*

Predefined functions means that each IO pin has a set of direct connections to certain signals from on-chip components. During run-time, the user can configure which component signal from a predefined set to connect to a certain pin at a certain time via memory mapped registers.

- **Analog pins** that have exclusively-dedicated **analog functions** – see Table 2-9 *Analog Pins*
- **Power pins** supply power to the chip components and non-power pins – see Table 2-10 *Power Pins*

Table 2-1 *Pin Overview* gives an overview of all the pins. For more information, see respective sections below. Alternatively, see [Appendix A – ESP32-S31 Consolidated Pin Overview](#).

Table 2-1. Pin Overview

Pin No.	Pin Name	Pin Type	Pin Providing Power ²	Pin Settings ³		Pin Function Sets ¹		
				At Reset	After Reset	IO MUX	LP IO MUX	Analog
1	ANT	Analog						
2	VDDA3	Power						
3	VDDA4	Power						
4	CHIP_PU	Analog						
5	XTAL_32K_N/GPIO0	IO	VDDPST_1			IO MUX	LP IO MUX	Analog
6	XTAL_32K_P/GPIO1	IO	VDDPST_1			IO MUX	LP IO MUX	Analog
7	GPIO2	IO	VDDPST_1			IO MUX	LP IO MUX	
8	GPIO3	IO	VDDPST_1			IO MUX	LP IO MUX	
9	GPIO4	IO	VDDPST_1			IO MUX	LP IO MUX	
10	GPIO5	IO	VDDPST_1			IO MUX	LP IO MUX	
11	VDDPST_1	Power						
12	GPIO6	IO	VDDPST_1			IO MUX	LP IO MUX	Analog
13	GPIO7	IO	VDDPST_1			IO MUX	LP IO MUX	Analog
14	GPIO8	IO	VDDPST_1		IE	IO MUX		Analog
15	GPIO9	IO	VDDPST_1		IE	IO MUX		Analog
16	GPIO10	IO	VDDPST_1		IE	IO MUX		Analog
17	GPIO11	IO	VDDPST_1		IE	IO MUX		Analog
18	VREF_TOUCH	Analog						
19	GPIO12	IO	VDDPST_1		IE	IO MUX		Analog
20	GPIO13	IO	VDDPST_1		IE	IO MUX		Analog

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Pin No.	Pin Name	Pin Type	Pin Providing Power ²	Pin Settings ³		Pin Function Sets ¹		
				At Reset	After Reset	IO MUX	LP IO MUX	Analog
21	GPIO14	IO	VDDPST_1		IE	IO MUX		Analog
22	GPIO15	IO	VDDPST_1		IE	IO MUX		Analog
23	GPIO16	IO	VDDPST_1		IE	IO MUX		Analog
24	GPIO17	IO	VDDPST_1		IE	IO MUX		Analog
25	GPIO18	IO	VDDPST_1		IE	IO MUX		Analog
26	GPIO19	IO	VDDPST_1		IE	IO MUX		Analog
27	SDIO_DATA0	IO	VDDPST_SD		IE	IO MUX		
28	SDIO_DATA1	IO	VDDPST_SD		IE	IO MUX		
29	SDIO_DATA2	IO	VDDPST_SD		IE	IO MUX		
30	VDD_PSRAM_1P8_1	Power						
31	SDIO_DATA3	IO	VDDPST_SD		IE	IO MUX		
32	SDIO_CLK	IO	VDDPST_SD		IE	IO MUX		
33	SDIO_CMD	IO	VDDPST_SD		IE	IO MUX		
34	VDD_PSRAM_1P8_2	Power						
35	VDD_LDO_1P8	Power						
36	SPICS	IO	VDD_SPI	WPU	WPU, IE	IO MUX		
37	SPIQ	IO	VDD_SPI	WPU	WPU, IE	IO MUX		
38	SPIWP	IO	VDD_SPI	WPU	WPU, IE	IO MUX		
39	VDD_SPI	Power						
40	SPIHD	IO	VDD_SPI	WPU	WPU, IE	IO MUX		
41	SPICLK	IO	VDD_SPI	WPU	WPU, IE	IO MUX		
42	SPID	IO	VDD_SPI	WPU	WPU, IE	IO MUX		
43	VCCA/VDDPST_2	Power						
44	USB_DP	Analog	VDDPST_2					
45	USB_DM	Analog	VDDPST_2					
46	GPIO33	IO	VDDPST_3	DRV=3	USB_PU, IE, DRV=3	IO MUX		Analog
47	GPIO34	IO	VDDPST_3	DRV=3	USB_PU, IE, DRV=3	IO MUX		Analog
48	GPIO35	IO	VDDPST_3		IE	IO MUX		
49	GPIO36	IO	VDDPST_3	IE	IE	IO MUX		
50	GPIO37	IO	VDDPST_3	IE	IE	IO MUX		Analog
51	GPIO38	IO	VDDPST_3	IE	IE	IO MUX		Analog
52	GPIO39	IO	VDDPST_3	IE	IE	IO MUX		Analog
53	GPIO40	IO	VDDPST_3	IE	IE	IO MUX		Analog
54	VDDPST_3	Power						
55	GPIO42	IO	VDDPST_3		IE	IO MUX		Analog
56	GPIO43	IO	VDDPST_3		IE	IO MUX		Analog
57	GPIO44	IO	VDDPST_3		IE	IO MUX		Analog
58	GPIO45	IO	VDDPST_3		IE	IO MUX		Analog
59	GPIO46	IO	VDDPST_3		IE	IO MUX		Analog
60	GPIO47	IO	VDDPST_3		IE	IO MUX		Analog
61	GPIO48	IO	VDDPST_3			IO MUX		Analog
62	GPIO49	IO	VDDPST_3			IO MUX		Analog
63	VREF_ADC	Analog						
64	VDDPST_4	Power						
65	GPIO50	IO	VDDPST_4			IO MUX		Analog

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Pin No.	Pin Name	Pin Type	Pin Providing Power ²	Pin Settings ³		Pin Function Sets ¹		
				At Reset	After Reset	IO MUX	LP IO MUX	Analog
66	GPIO51	IO	VDDPST_4			IO MUX		Analog
67	GPIO52	IO	VDDPST_4		IE	IO MUX		Analog
68	GPIO53	IO	VDDPST_4		IE	IO MUX		Analog
69	MTDO	IO	VDDPST_4		IE	IO MUX		Analog
70	MTCK	IO	VDDPST_4		IE	IO MUX		Analog
71	MTDI	IO	VDDPST_4		IE	IO MUX		Analog
72	MTMS	IO	VDDPST_4		IE	IO MUX		Analog
73	GPIO58	IO	VDDPST_4		IE	IO MUX		
74	GPIO59	IO	VDDPST_4		IE	IO MUX		
75	GPIO60	IO	VDDPST_4	WPU, IE	WPU, IE	IO MUX		
76	GPIO61	IO	VDDPST_4	WPU, IE	WPU, IE	IO MUX		
77	VDDA1	Power						
78	XTAL_N	Analog						
79	XTAL_P	Analog						
80	VDDA2	Power						
81	GND	Power						

- Bold** marks the pin function set in which a pin has its default function in the default boot mode. For more information about the default boot mode, see Section 3.1 [Chip Boot Mode Control](#).
- In column **Pin Providing Power**:
 - Pins powered by VDD_SPI: The actual power source is the internal power rail that supplies VDD_SPI. For details, see Section 2.5.2 [Power Scheme](#).
 - Pins powered by VDDPST_SD: VDDPST_SD is an internal SDIO IO power domain, not a package power pin. The power source can be VDDPST_2 or VDD_LDO_1P8, configurable via internal registers. For details, see Section 2.5.2 [Power Scheme](#).
- Column **Pin Settings** shows predefined settings at reset and after reset with the following abbreviations:
 - IE – input enabled
 - WPU – internal weak pull-up resistor enabled
 - WPD – internal weak pull-down resistor enabled
 - DRV – drive strength
 - Values 0~3 correspond to ~5, ~10, ~20, and ~40 mA, respectively.
 - Default drive current is 40 mA for GPIO33 and GPIO34, and 20 mA for other IO pins.
 - USB_PU – USB pull-up resistor enabled
 - By default, the USB function is enabled for USB pins (i.e., GPIO33 and GPIO34), and the pin pull-up is decided by the USB pull-up. The USB pull-up is controlled by USB_SERIAL_JTAG_DP/DM_PULLUP and the pull-up resistor value is controlled by USB_SERIAL_JTAG_PULLUP_VALUE.
 - When the USB function is disabled, USB pins are used as regular GPIOs and the pin's internal weak pull-up and pull-down resistors are disabled by default (configurable by IO_MUX_FUN_WPU/WPD).

2.3 IO Pins

2.3.1 IO MUX Functions

The IO MUX allows multiple input/output signals to be connected to a single input/output pin. Each IO pin of ESP32-S31 can be connected to one of the five signals (IO MUX functions, i.e., F0–F4), as listed in Table 2-3 *IO MUX Functions*.

Among the five sets of signals:

- Some are routed via the GPIO Matrix (**GPIO0, GPIO1, etc.**), which incorporates internal signal routing circuitry for mapping signals programmatically. It gives the pin access to almost any peripheral signals. However, the flexibility of programmatic mapping comes at a cost as it might affect the latency of routed signals.
- Some are directly routed from certain peripherals, including UART0, JTAG, SPI0/1, and SPI2 signals - see Table 2-2 *Peripheral Signals Routed via IO MUX*.

Table 2-2. Peripheral Signals Routed via IO MUX

Pin Function	Signal	Description
MTCK MTDO MTDI MTMS	Test clock Test data out Test data in Test mode select	JTAG interface for debugging
SPICS SPIQ SPIWP SPIHD SPICLK SPID	Chip select Master in, slave out Write protect Hold Clock Master out, slave in	3.3 V SPI0/1 interface for connection to off-package flash via the SPI bus. It supports 1-line, 2-line, and 4-line SPI modes. See also Section 2.6 <i>Pin Mapping Between Chip and Flash</i>
SPI2_HOLD_PAD SPI2_CS_PAD SPI2_D_PAD SPI2_CK_PAD SPI2_Q_PAD SPI2_WP_PAD	Hold Chip select Data in Clock Data out Write protect	3.3 V SPI2 interface which can operate in master and slave modes. The interface supports 1-line, 2-line, 4-line, and 8-line modes (the 8-line mode is supported only in the master mode).
SPI2_IO..._PAD SPI2_DQS_PAD	Data Data strobe/data mask	The high 4-bit data line interface and the DQS interface for 3.3 V SPI2 interface in 8-line SPI mode
UART0_TXD_PAD UART0_RXD_PAD	Transmit data Receive data	UART0 Interface

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Table 2-2 – cont'd from previous page

Pin Function	Signal	Description
GMAC_PHY_RXDV_PAD ¹	Receive data valid	Gigabit Ethernet interface
GMAC_PHY_RXD..._PAD	Receive data line 0/1	
GMAC_RX_CLK_PAD	Receive clock	
GMAC_PHY_TXDV_PAD	Transmit data valid	
GMAC_PHY_TXD..._PAD	Transmit data line 0/1	
GMAC_PHY_TXEN_PAD	Transmit enable	
GMAC_RMII_CLK_PAD	RMII clock	
REF_GMAC_CLK_PAD	PHY PLL reference clock	
SDIO_DATA...	Data line 0–3 of card one	SD/SDIO/MMC host controller
SDIO_CLK	Clock of card one	
SDIO_CMD	Command of card one	
SD2_CDATA..._PAD	Data line 0–3 of card two	
SD2_CCLK_PAD	Clock of card two	
SD2_CCMD_PAD	Command of card two	
CAM_DATA..._PAD	Data in	Camera interface
CAM_PCLK_PAD	Pixel clock in	
CAM_XCLK_PAD	Clock out	
CAM_V_SYNC_PAD	Vertical sync	
CAM_H_SYNC_PAD	Horizontal sync	
LCD_DATA..._OUT_PAD	Data out	LCD display interface
LCD_PCLK_PAD	Pixel clock out	
LCD_H_ENABLE_PAD	Horizontal data enable	
LCD_V_SYNC_PAD	Vertical sync	
LCD_H_SYNC_PAD	Horizontal sync	

¹ The PAD layer does not distinguish between MII and RMII interfaces. This signal is used as RX_DV in MII mode and as CRS_DV in RMII mode.

Table 2-3 *IO MUX Functions* shows the IO MUX functions of IO pins.

Table 2-3. IO MUX Functions

Pin No.	GPIO	IO MUX Functions									
		F0	Type	F1	Type	F2	Type	F3	Type	F4	Type
5	XTAL_32K_N	GPIO0	I/O/T	GPIO0	I/O/T						
6	XTAL_32K_P	GPIO1	I/O/T	GPIO1	I/O/T						
7	GPIO2	GPIO2	I/O/T	GPIO2	I/O/T			LCD_DATA19_OUT_PAD	O		
8	GPIO3	GPIO3	I/O/T	GPIO3	I/O/T			LCD_DATA20_OUT_PAD	O		
9	GPIO4	GPIO4	I/O/T	GPIO4	I/O/T			LCD_DATA21_OUT_PAD	O		
10	GPIO5	GPIO5	I/O/T	GPIO5	I/O/T			LCD_DATA22_OUT_PAD	O		
12	GPIO6	GPIO6	I/O/T	GPIO6	I/O/T						
13	GPIO7	GPIO7	I/O/T	GPIO7	I/O/T			LCD_DATA23_OUT_PAD	O		
14	GPIO8	GPIO8	I/O/T	GPIO8	I/O/T	GMAC_PHY_TXD0_PAD	O	LCD_DATA0_OUT_PAD	O		
15	GPIO9	SPI2_HOLD_PAD	I1/O/T	GPIO9	I/O/T	GMAC_PHY_TXD1_PAD	O	LCD_DATA1_OUT_PAD	O		
16	GPIO10	SPI2_CS_PAD	I1/O/T	GPIO10	I/O/T	GMAC_PHY_TXD2_PAD	O	LCD_DATA2_OUT_PAD	O		
17	GPIO11	SPI2_D_PAD	I1/O/T	GPIO11	I/O/T	GMAC_PHY_TXD3_PAD	O	LCD_DATA3_OUT_PAD	O		
19	GPIO12	SPI2_CK_PAD	I1/O/T	GPIO12	I/O/T	GMAC_PHY_TXEN_PAD	O	LCD_DATA4_OUT_PAD	O		
20	GPIO13	SPI2_Q_PAD	I1/O/T	GPIO13	I/O/T	GMAC_RMII_CLK_PAD	IO/O/T	LCD_DATA5_OUT_PAD	O		
21	GPIO14	SPI2_WP_PAD	I1/O/T	GPIO14	I/O/T	GMAC_RX_CLK_PAD	IO	LCD_DATA6_OUT_PAD	O		
22	GPIO15	SPI2_IO4_PAD	I1/O/T	GPIO15	I/O/T	GMAC_PHY_RXDV_PAD	IO	LCD_DATA7_OUT_PAD	O		
23	GPIO16	SPI2_IO5_PAD	I1/O/T	GPIO16	I/O/T	GMAC_PHY_RXD3_PAD	IO	LCD_DATA8_OUT_PAD	O		
24	GPIO17	SPI2_IO6_PAD	I1/O/T	GPIO17	I/O/T	GMAC_PHY_RXD2_PAD	IO	LCD_DATA9_OUT_PAD	O		
25	GPIO18	SPI2_IO7_PAD	I1/O/T	GPIO18	I/O/T	GMAC_PHY_RXD1_PAD	IO	LCD_DATA10_OUT_PAD	O		
26	GPIO19	SPI2_DQS_PAD	O/T	GPIO19	I/O/T	GMAC_PHY_RXD0_PAD	IO	LCD_DATA11_OUT_PAD	O		
27	SDIO_DATA0	SDIO_DATA0	I1/O/T	GPIO20	I/O/T	SPI2_CK_PAD	I1/O/T				
28	SDIO_DATA1	SDIO_DATA1	I1/O/T	GPIO21	I/O/T	SPI2_D_PAD	I1/O/T				
29	SDIO_DATA2	SDIO_DATA2	I1/O/T	GPIO22	I/O/T	SPI2_Q_PAD	I1/O/T				
31	SDIO_DATA3	SDIO_DATA3	I1/O/T	GPIO23	I/O/T	SPI2_CS_PAD	I1/O/T				
32	SDIO_CLK	SDIO_CLK	O	GPIO24	I/O/T	SPI2_HOLD_PAD	I1/O/T				
33	SDIO_CMD	SDIO_CMD	I1/O/T	GPIO25	I/O/T	SPI2_WP_PAD	I1/O/T				
36	SPICS	SPICS	O/T	GPIO26	I/O/T						
37	SPIQ	SPIQ	I1/O/T	GPIO27	I/O/T						

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Pin No.	GPIO	IO MUX Functions									
		F0	Type	F1	Type	F2	Type	F3	Type	F4	Type
38	SPIWP	SPIWP	I/O/T	GPIO28	I/O/T						
40	SPIHD	SPIHD	I/O/T	GPIO30	I/O/T						
41	SPICLK	SPICLK	O/T	GPIO31	I/O/T						
42	SPID	SPID	I/O/T	GPIO32	I/O/T						
46	GPIO33	GPIO33	I/O/T	GPIO33	I/O/T			LCD_DATA12_OUT_PAD	O		
47	GPIO34	GPIO34	I/O/T	GPIO34	I/O/T			LCD_DATA13_OUT_PAD	O		
48	GPIO35	GPIO35	I/O/T	GPIO35	I/O/T	REF_GMAC_CLK_PAD	O	LCD_DATA14_OUT_PAD	O	SD2_CDATA0_PAD	I/O/T
49	GPIO36	GPIO36	I/O/T	GPIO36	I/O/T			LCD_DATA15_OUT_PAD	O	SD2_CDATA1_PAD	I/O/T
50	GPIO37	GPIO37	I/O/T	GPIO37	I/O/T			LCD_DATA16_OUT_PAD	O	SD2_CDATA2_PAD	I/O/T
51	GPIO38	GPIO38	I/O/T	GPIO38	I/O/T			LCD_DATA17_OUT_PAD	O	SD2_CDATA3_PAD	I/O/T
52	GPIO39	GPIO39	I/O/T	GPIO39	I/O/T			LCD_DATA18_OUT_PAD	O	SD2_CCLK_PAD	O
53	GPIO40	GPIO40	I/O/T	GPIO40	I/O/T			LCD_PCLK_PAD	O	SD2_CCMD_PAD	I/O/T
55	GPIO42	GPIO42	I/O/T	GPIO42	I/O/T						
56	GPIO43	GPIO43	I/O/T	GPIO43	I/O/T			LCD_H_ENABLE_PAD	O		
57	GPIO44	GPIO44	I/O/T	GPIO44	I/O/T			LCD_H_SYNC_PAD	O		
58	GPIO45	GPIO45	I/O/T	GPIO45	I/O/T			LCD_V_SYNC_PAD	O		
59	GPIO46	GPIO46	I/O/T	GPIO46	I/O/T			CAM_DATA0_IN_PAD	IO		
60	GPIO47	GPIO47	I/O/T	GPIO47	I/O/T			CAM_DATA1_IN_PAD	IO		
61	GPIO48	GPIO48	I/O/T	GPIO48	I/O/T			CAM_DATA2_IN_PAD	IO		
62	GPIO49	GPIO49	I/O/T	GPIO49	I/O/T			CAM_DATA3_IN_PAD	IO		
65	GPIO50	GPIO50	I/O/T	GPIO50	I/O/T			CAM_DATA4_IN_PAD	IO		
66	GPIO51	GPIO51	I/O/T	GPIO51	I/O/T			CAM_DATA5_IN_PAD	IO		
67	GPIO52	GPIO52	I/O/T	GPIO52	I/O/T	SPI2_CS_PAD	I/O/T	CAM_DATA6_IN_PAD	IO		
68	GPIO53	GPIO53	I/O/T	GPIO53	I/O/T	SPI2_CK_PAD	I/O/T	CAM_DATA7_IN_PAD	IO		
69	MTDO	MTDO	O/T	GPIO54	I/O/T	SPI2_D_PAD	I/O/T	CAM_PCLK_PAD	IO		
70	MTCK	MTCK	I	GPIO55	I/O/T	SPI2_Q_PAD	I/O/T	CAM_XCLK_PAD	O		
71	MTDI	MTDI	I	GPIO56	I/O/T	SPI2_HOLD_PAD	I/O/T	CAM_V_SYNC_PAD	IO		
72	MTMS	MTMS	I	GPIO57	I/O/T	SPI2_WP_PAD	I/O/T	CAM_H_SYNC_PAD	IO		
73	GPIO58	UART0_TXD_PAD	O	GPIO58	I/O/T						

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Pin No.	GPIO	IO MUX Functions									
		F0	Type	F1	Type	F2	Type	F3	Type	F4	Type
74	GPIO59	UART0_RXD_PAD	I1	GPIO59	I/O/T						
75	GPIO60	GPIO60	I/O/T	GPIO60	I/O/T						
76	GPIO61	GPIO61	I/O/T	GPIO61	I/O/T						

¹ **Bold** marks the default pin functions in the default boot mode. See Section [3.1 Chip Boot Mode Control](#).

² Regarding **red** or **yellow** highlighted cells, see Section [2.3.4 Restrictions for GPIOs and LP GPIOs](#).

³ Each IO MUX function (F_n , $n = 0-4$) is associated with a *type*. The description of *type* is as follows:

- I – input. O – output. T – high impedance.
- I1 – input; if the pin is assigned a function other than F_n , the input signal of F_n is always 1.
- IO – input; if the pin is assigned a function other than F_n , the input signal of F_n is always 0.

2.3.2 LP IO MUX Functions

When the chip is in Deep-sleep mode, the IO MUX described in Section 2.3.1 *IO MUX Functions* is not available. This is where the LP IO MUX is used. It allows multiple input/output signals to be a single input/output pin in Deep-sleep mode, as the pin is connected to the LP system and powered by VDDPST_1.

LP IO pins can be assigned to LP functions. They can:

- Either work as LP GPIOs (**LP_GPIO0, LP_GPIO1, etc.**), connected to the LP CPU
- Or connect to LP peripheral signals (**LP_I2C_SDA_PAD, LP_I2C_SCL_PAD, etc.**) – see Table 2-4 *LP Peripheral Signals Routed via LP IO MUX*

Table 2-4. LP Peripheral Signals Routed via LP IO MUX

Pin Function	Signal	Description
LP_I2C_SDA_PAD	Serial data	LP I2C interface
LP_I2C_SCL_PAD	Serial clock	LP I2C interface
LP_UART_RXD_PAD	Receive	LP UART interface
LP_UART_TXD_PAD	Transmit	LP UART interface
LP_UART_RTSN_PAD	Request to send	LP UART interface
LP_UART_CTSN_PAD	Clear to send	LP UART interface
LP_UART_DTRN_PAD	Data terminal ready	LP UART interface
LP_UART_DSRN_PAD	Data set ready	LP UART interface
LP_SPI_CK_PAD	Clock	LP SPI interface
LP_SPI_CS_PAD	Chip select	LP SPI interface
LP_SPI_D_PAD	Master out, slave in	LP SPI interface
LP_SPI_Q_PAD	Master in, slave out	LP SPI interface

Table 2-5 *LP IO MUX Functions* lists the LP IO MUX functions of LP IO pins.

Table 2-5. LP IO MUX Functions

Pin No.	LP IO Name ¹	LP IO MUX Function							
		F0	Type	F1	Type	F2	Type	F3	Type
5	LP_GPIO0	LP_GPIO0	I/O/T	LP_GPIO0	I/O/T			LP_PROBE_TOP_OUT0	O
6	LP_GPIO1	LP_GPIO1	I/O/T	LP_GPIO1	I/O/T			LP_PROBE_TOP_OUT1	O
7	LP_GPIO2	LP_UART_DTRN_PAD	O	LP_GPIO2	I/O/T	LP_SPI_CK_PAD	I/O/T	LP_PROBE_TOP_OUT2	O
8	LP_GPIO3	LP_UART_DSRN_PAD	I1	LP_GPIO3	I/O/T	LP_SPI_CS_PAD	I/O/T	LP_PROBE_TOP_OUT3	O
9	LP_GPIO4	LP_UART_RTSN_PAD	O	LP_GPIO4	I/O/T	LP_SPI_D_PAD	I/O/T	LP_PROBE_TOP_OUT4	O
10	LP_GPIO5	LP_UART_CTSN_PAD	I1	LP_GPIO5	I/O/T	LP_SPI_Q_PAD	I/O/T	LP_PROBE_TOP_OUT5	O
12	LP_GPIO6	LP_UART_TXD_PAD	O	LP_GPIO6	I/O/T	LP_I2C_SCL_PAD	I/O/T	LP_PROBE_TOP_OUT6	O
13	LP_GPIO7	LP_UART_RXD_PAD	I1	LP_GPIO7	I/O/T	LP_I2C_SDA_PAD	I/O/T	LP_PROBE_TOP_OUT7	O

¹ This column lists LP GPIO names because LP IO MUX functions are configured via LP GPIO registers that use LP GPIO numbering.

² LP UART hardware flow control and LP I2C cannot be used at the same time.

2.3.3 Analog Functions

Some IO pins also have **analog functions**, for analog peripherals (such as ADC) in any power mode. Internal analog signals are routed to these analog functions, see Table 2-6 *Analog Signals Routed to Analog Functions*.

Table 2-6. Analog Signals Routed to Analog Functions

Pin Function	Signal	Description
TOUCH_CH...	Touch sensor channel ... signal	Touch sensor interface
ADC..._CH...	ADC1/2 channel ... signal	ADC1/2 interface
XTAL_32K_N XTAL_32K_P	Negative clock signal Positive clock signal	External 32 kHz clock input/output for ESP32-S31 passive or active crystal oscillator
USB1P1_N... USB1P1_P...	USB D- USB D+	USB Serial/JTAG function
PAD_COMP...	Analog voltage comparator channel... signal	Analog voltage comparator interface

Table 2-7 *Analog Functions* shows the analog functions of IO pins.

Table 2-7. Analog Functions

Pin No.	Analog IO Name	Analog Functions FO
5	GPIO0	XTAL_32K_N
6	GPIO1	XTAL_32K_P
12	GPIO6	TOUCH_CH0
13	GPIO7	TOUCH_CH1
14	GPIO8	TOUCH_CH2
15	GPIO9	TOUCH_CH3
16	GPIO10	TOUCH_CH4
17	GPIO11	TOUCH_CH5
19	GPIO12	TOUCH_CH6
20	GPIO13	TOUCH_CH7
21	GPIO14	TOUCH_CH8
22	GPIO15	TOUCH_CH9
23	GPIO16	TOUCH_CH10
24	GPIO17	TOUCH_CH11
25	GPIO18	TOUCH_CH12
26	GPIO19	TOUCH_CH13
46	GPIO33	USB1P1_NO
47	GPIO34	USB1P1_PO
50	GPIO37	PAD_COMP0
51	GPIO38	PAD_COMP1
52	GPIO39	PAD_COMP2
53	GPIO40	PAD_COMP3
55	GPIO42	ADC1_CH0_N

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Pin No.	Analog IO Name	Analog Functions FO
56	GPIO43	ADC1_CH0_P
57	GPIO44	ADC1_CH1_N
58	GPIO45	ADC1_CH1_P
59	GPIO46	ADC1_CH2_N
60	GPIO47	ADC1_CH2_P
61	GPIO48	ADC1_CH3_N
62	GPIO49	ADC1_CH3_P
65	GPIO50	ADC2_CH0_N
66	GPIO51	ADC2_CH0_P
67	GPIO52	ADC2_CH1_N
68	GPIO53	ADC2_CH1_P
69	GPIO54	ADC2_CH2_N
70	GPIO55	ADC2_CH2_P
71	GPIO56	ADC2_CH3_N
72	GPIO57	ADC2_CH3_P

¹ **Bold** marks the default pin functions in the default boot mode. See Section [3.1 Chip Boot Mode Control](#).

² This column lists analog IO names. Since analog functions are configured with GPIO registers that use GPIO numbering, GPIO names are usually listed.

³ Regarding **highlighted** cells, see Section [2.3.4 Restrictions for GPIOs and LP GPIOs](#).

2.3.4 Restrictions for GPIOs and LP GPIOs

All IO pins of the ESP32-S31 have GPIO and some have LP GPIO pin functions. However, the IO pins are multiplexed and can be configured for different purposes based on the requirements. Some IOs have restrictions for usage. It is essential to consider the multiplexed nature and the limitations when using these IO pins.

In tables of this chapter, some pin functions are **highlighted in red** or **highlighted in yellow**. The non-highlighted GPIO or LP GPIO pins are recommended for use first. If more pins are needed, the highlighted GPIOs or LP GPIOs should be chosen carefully to avoid conflicts with important pin functions.

The highlighted IO pins have the following important pin functions:

- **GPIO** – allocated for communication with flash/PSRAM and NOT recommended for other uses. For details, see Section [2.6 Pin Mapping Between Chip and Flash](#).
- **GPIO** – have one of the following important functions:
 - **Strapping pins** – need to be at certain logic levels at startup. See Section [3 Boot Configurations](#).

Note:

Strapping pins are highlighted by pin name, instead of pin functions.

- **USB1P1_P/N** – by default, connected to the USB Serial/JTAG controller. To function as GPIOs, these pins need to be reconfigured.
- **JTAG interface** – often used for debugging. See Table [2-2 Peripheral Signals Routed via IO MUX](#). To free these pins up, the pin functions USB1P1_N/P of the USB Serial/JTAG controller can be used instead. See also Section [3.4 JTAG Signal Source Control](#).
- **UART interface** – often used for debugging. See Table [2-2 Peripheral Signals Routed via IO MUX](#).

See also [Appendix A – ESP32-S31 Consolidated Pin Overview](#).

2.3.5 Peripheral Pin Assignment

Table 2-8 *Peripheral Pin Assignment* highlights which pins can be assigned to each peripheral interface according to the following priorities:

- **Priority 1 (P1)** : Fixed pins connected directly to peripheral signals via IO MUX or LP IO MUX.
If a peripheral interface does not have priority 1 pins, such as UART2, it can be assigned to any GPIO pins from priority 2 to priority 4.
- Any GPIO or LP GPIO pins mapping to peripheral signals via GPIO Matrix or LP GPIO Matrix can be priority 2, 3, or 4.
 - **Priority 2 (P2)** : GPIO pins can be freely used without restrictions.
 - **Priority 3 (P3)** : GPIO pins should be used with caution, as they may conflict with the following important functions described in Section 2.3.4 *Restrictions for GPIOs and LP GPIOs*:
 - * GPIO36, GPIO37, GPIO60, GPIO61 : Strapping pins.
 - * GPIO33, GPIO34 : USB Serial/JTAG interface.
 - * GPIO54, GPIO55, GPIO56, GPIO57 : JTAG interface.
 - * GPIO58, GPIO59 : UART0 interface.
 - **Priority 4 (P4)** : GPIO pins already allocated or not recommended for use, as described in Section 2.3.4 *Restrictions for GPIOs and LP GPIOs*:
 - * GPIO26, GPIO27, GPIO28, GPIO30, GPIO31, GPIO32 : SPI0/1 interface pins recommended for connecting off-package flash.

If a peripheral interface does not have priority 2 to 4 pins, such as USB Serial/JTAG, it means it can be assigned only to priority 1 pins.

Note:

- For details about which peripheral signals are connected to IO MUX or LP IO MUX pins, please refer to Section 2.3.1 *IO MUX Functions* or Section 2.3.2 *LP IO MUX Functions*.

Table 2-8. Peripheral Pin Assignment

[illegible]

¹ For USB Serial/JTAG, use USB1P1_N0 and USB1P1_P0. For USB OTG, use USB_D- and USB_D+ for USB 2.0 High-Speed OTG PHY signals, while other USB OTG signals can be routed through the GPIO Matrix.

² Signals of UART0, LP UART, LP SPI, LP I2C, SPI2, EMAC, and LCD and Camera interfaces can be mapped to available pins through the GPIO Matrix or LP GPIO Matrix, regardless of whether they are directly routed to fixed pins via IO MUX or LP IO MUX. LP UART hardware flow control and LP I2C cannot be used at the same time. For details, see Table 2-5 LP IO MUX Functions.

2.4 Analog Pins

Table 2-9. Analog Pins

Pin No.	Pin Name	Pin Type	Pin Function
1	ANT	I/O	RF input and output
4	CHIP_PU	I	High level: Chip enabled (powered on); Low level: Chip disabled (powered off); Do not leave CHIP_PU floating
18	VREF_TOUCH	I/O	TOUCH reference voltage
44	USB_DP	I/O	USB 2.0 High-Speed OTG PHY D+
45	USB_DM	I/O	USB 2.0 High-Speed OTG PHY D-
63	VREF_ADC	I/O	ADC reference voltage
78	XTAL_N	—	External clock input/output for connecting active or passive crystal oscillators of ESP32-S31. For an active crystal oscillator, connect to P and leave N floating
79	XTAL_P		

2.5 Power Supply

2.5.1 Power Pins

The chip is powered via the power pins described in Table 2-10 *Power Pins*.

Table 2-10. Power Pins

Pin No.	Pin Name	Direction	Power Supply ¹	
			Power Domain / Other ²	IO Pins
2	VDDA3	Input	Analog power domain	
3	VDDA4	Input	Analog power domain	
11	VDDPST_1	Input	Digital LP power domain	LP IO ²
30	VDD_PSRAM_1P8_1	Input	PSRAM	PSRAM IO
34	VDD_PSRAM_1P8_2	Input	PSRAM	PSRAM IO
35	VDD_LDO_1P8	Output	PSRAM/FLASH/SD	PSRAM/FLASH/SD IO
39	VDD_SPI	Input	FLASH	FLASH IO
		Output	FLASH	FLASH IO
43	VCCA/VDDPST_2	Input	USB_PHY/Digital HP power domain	USB IO
54	VDDPST_3	Input	Digital HP power domain	HP IO
64	VDDPST_4	Input	Digital HP power domain	HP IO
77	VDDA1	Input	Analog power domain	
80	VDDA2	Input	Analog power domain	
81	GND	–	External ground connection	

¹ See in conjunction with Section 2.5.2 *Power Scheme*.

² LP IO pins are those powered by VDDPST_1, as shown in Figure 2-2 *ESP32-S31 Power Scheme*. See also Table 2-1 *Pin Overview* > Column *Pin Providing Power*.

2.5.2 Power Scheme

The power scheme is shown in Figure 2-2 *ESP32-S31 Power Scheme*.

The components on the chip are powered via voltage regulators.

Table 2-11. Voltage Regulators

Voltage Regulator	Output	Power Supply
HP_LDO+MEM_LDO+MEM_LDO_SLV	1.1 V	Digital power domain
LP_LDO	1.1 V	LP power domain
PSRAM	1.8 V	Configurable to supply in-package PSRAM or off-package memory
FLASH	3.3 V	Configurable to supply in-package flash
SDIO	3.3 V/1.8 V	Configurable as the SDIO power supply

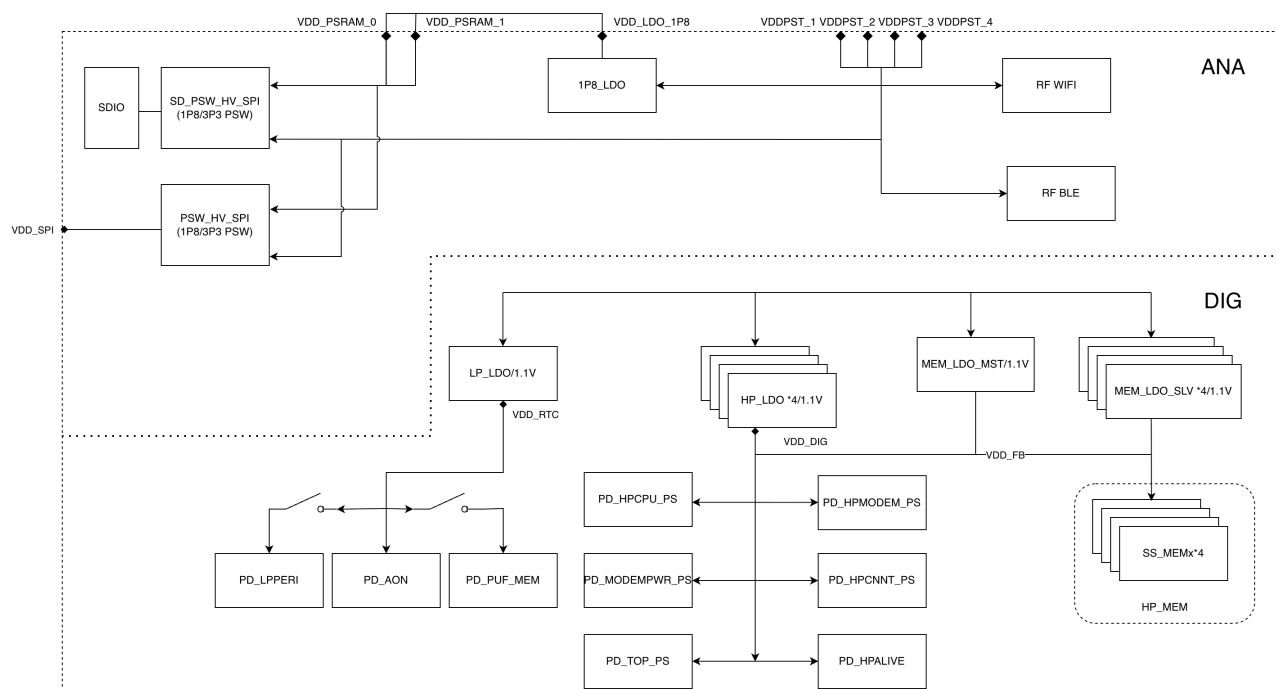


Figure 2-2. ESP32-S31 Power Scheme

2.5.3 Chip Power-up and Reset

Once the power is supplied to the chip, its power rails need a short time to stabilize. After that, CHIP_PU – the pin used for power-up and reset – is pulled high to activate the chip. For information on CHIP_PU as well as power-up and reset timing, see Figure 2-3 and Table 2-12.

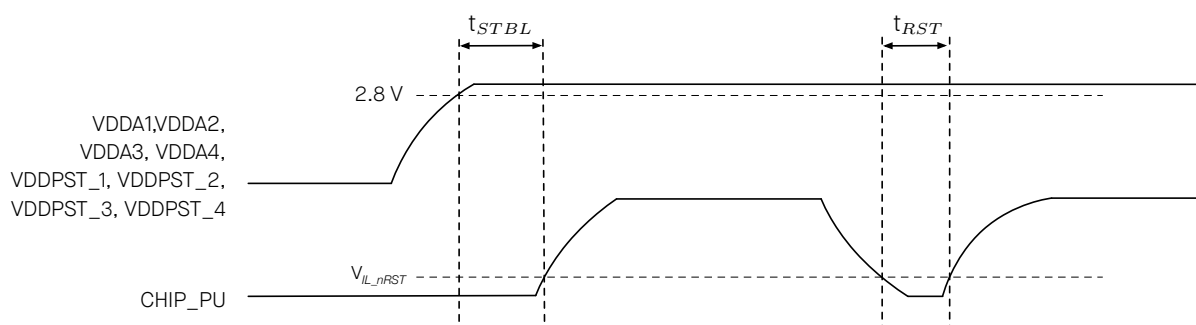


Figure 2-3. Visualization of Timing Parameters for Power-up and Reset

Table 2-12. Description of Timing Parameters for Power-up and Reset

Parameter	Description	Min (ms)
t_{STBL}	Time reserved for the power rails of VDDA1, VDDA2, VDDA3, VDDA4, VDDPST_1, VDDPST_2, VDDPST_3, VDDPST_4 to stabilize before the CHIP_PU pin is pulled high to activate the chip	1
t_{RST}	Time reserved for CHIP_PU to stay below V_{IL_nRST} to reset the chip (see Table 5-4)	1

2.6 Pin Mapping Between Chip and Flash

ESP32-S31 requires off-package flash to store application firmware and data. ESP32-S31 supports up to 256 MB flash, which can be connected through SPI, Dual SPI, and Quad SPI/QPI.

ESP32-S31 includes eight-line PSRAM with the operation voltage of 1.8 V. Please note that PSRAM is not pinned out.

Table 2-13 lists the pin mapping between the chip and flash for all SPI modes.

For more information on SPI controllers, see also Section 4.2.2.2 *SPI Controller (SPI)*.

Table 2-13. Pin Mapping Between Chip and Off-Package Flash

Pin No.	Pin Name	Single SPI	Dual SPI	Quad SPI/QPI
36	SPICS	CS#	CS#	CS#
37	SPIQ	DO	DO	DO
38	SPIWP	WP#	WP#	WP#
40	SPIHD	HOLD#	HOLD#	HOLD#
41	SPICLK	CLK	CLK	CLK
42	SPID	DI	DI	DI

3 Boot Configurations

The chip allows for configuring the following boot parameters through [strapping pins](#) and [eFuse parameters](#) at power-up or a hardware reset, without microcontroller interaction.

- **Chip boot mode**
 - Strapping pin: GPIO60 and GPIO61
- **VDD_SPI voltage**
 - Strapping pin: GPIO36
 - eFuse parameter: EFUSE_PMU_FLASH_POWER_SEL and EFUSE_PMU_FLASH_POWER_SEL_EN
- **ROM message printing**
 - Strapping pin: GPIO60
 - eFuse parameter: EFUSE_UART_PRINT_CONTROL and EFUSE_DIS_USB_SERIAL_JTAG_ROM_PRINT
- **JTAG signal source**
 - Strapping pin: GPIO37
 - eFuse parameter: EFUSE_DIS_PAD_JTAG, EFUSE_DIS_USB_JTAG, and EFUSE_JTAG_SEL_ENABLE

The default values of all the above eFuse parameters are 0, which means that they are not burnt. Given that eFuse is one-time programmable, once programmed to 1, it can never be reverted to 0.

The default values of the strapping pins, namely the logic levels, are determined by the pins' internal weak pull-up/pull-down resistors at reset if the pins are not connected to any circuit, or connected to an external high-impedance circuit.

Table 3-1. Default Configuration of Strapping Pins

Strapping Pin	Default Configuration	Bit Value
GPIO36	Floating	–
GPIO37	Floating	–
GPIO60	Weak pull-up	1
GPIO61	Weak pull-up	1

To change the bit values, the strapping pins should be connected to external pull-down/pull-up resistances. If the ESP32-S31 is used as a device by a host MCU, the strapping pin voltage levels can also be controlled by the host MCU.

All strapping pins have latches. At Chip Reset, the latches sample the bit values of their respective strapping pins and store them until the chip is powered down or shut down. The states of latches cannot be changed in any other way. It makes the strapping pin values available during the entire chip operation, and the pins are freed up to be used as regular IO pins after reset.

The timing of signals connected to the strapping pins should adhere to the *setup time* and *hold time* specifications in [Table 3-2](#) and [Figure 3-1](#).

Table 3-2. Description of Timing Parameters for the Strapping Pins

Parameter	Description	Min (ms)
t_{SU}	Setup time is the time reserved for the power rails to stabilize before the CHIP_PU pin is pulled high to activate the chip.	0
t_H	Hold time is the time reserved for the chip to read the strapping pin values after CHIP_PU is already high and before these pins start operating as regular IO pins.	3

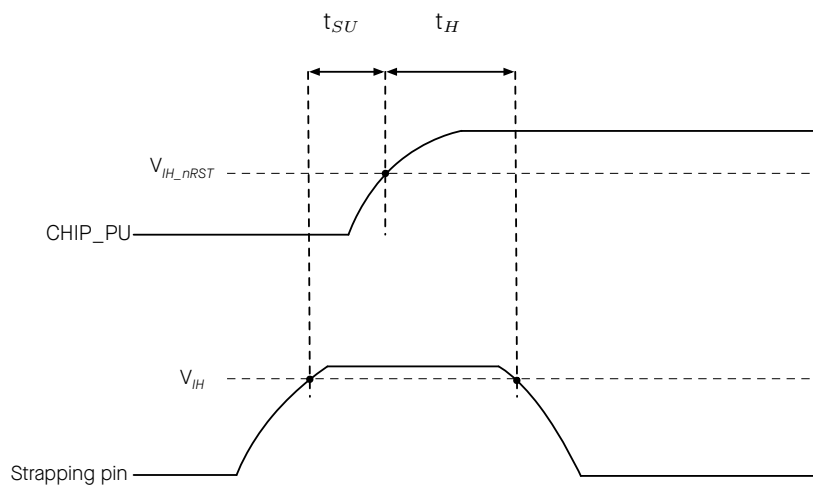


Figure 3-1. Visualization of Timing Parameters for the Strapping Pins

3.1 Chip Boot Mode Control

GPIO60 and GPIO61 control the boot mode after the reset is released. See Table 3-3 *Chip Boot Mode Control*.

Table 3-3. Chip Boot Mode Control

Boot Mode	GPIO61	GPIO60
SPI boot mode ¹	1	Any value
Joint download boot mode ²	0	1
Invalid combination ³	0	0

¹ **Bold** marks the default value and configuration.

² Joint Download Boot mode supports the following download methods:

- USB-Serial-JTAG Download Boot
- USB-OTG Download Boot
- UART Download Boot
- GPSPI Download Boot

³ This combination triggers unexpected behavior and should be avoided.

3.1.1 Secure Debugging in Joint Download Boot Mode

ESP32-S31 is the first chip to support the Secure Debug Controller (SDC).

When all of the following conditions are met, the chip supports SDC-related commands in Joint Download Boot mode:

- The chip has entered [Joint download boot mode](#)
- eFuse bit EFUSE_RMA_ENA has been burned

Note: SDC-related capabilities are not affected by eFuse bits EFUSE_DIS_DOWNLOAD_MODE and EFUSE_ENABLE_SECURE_DOWNLOAD.

Before using SDC, complete eFuse and certificate configuration as follows:

1. Burn the SDC public-key hash into eFuse;
2. In Joint Download Boot mode, transfer the SDC certificate to the chip via esptool; the ROM verifies the certificate in hardware.

After successful certificate verification, the ROM temporarily enables the following debugging capabilities at the hardware level (even if the corresponding eFuse bits have been burned to disable them):

- Software JTAG, even if EFUSE_SOFT_DIS_JTAG has been burned;
- Download mode, even if EFUSE_DIS_DOWNLOAD_MODE has been burned;
- Forced entry into SPI Boot mode for application software debugging.

The SDC authorization state is valid only for the current run. After power-on reset, the verification result is cleared; to debug again, repeat certificate transfer and verification.

3.2 VDD_SPI Voltage Control

VDD_SPI can be sourced from either the internal or external power supply:

- **(Default)** If register PMU_ACTIVE_VDD_SPI_PD_EN is low, power is supplied by the internal POWER_SWITCH
- If register PMU_ACTIVE_VDD_SPI_PD_EN is high, power is supplied externally

The actual VDD_SPI supply voltage level is jointly determined by EFUSE_PMU_FLASH_POWER_SEL, register PMU_ACTIVE_VDD_SPI_PD_EN, GPIO36, and EFUSE_PMU_FLASH_POWER_SEL_EN.

Table 3-4. VDD_SPI Voltage Control

Voltage Level ¹	EFUSE_PMU_FLASH_POWER_SEL	GPIO36	EFUSE_PMU_FLASH_POWER_SEL_EN
3.3 V	0	1	Ignored
	1	Ignored	1
1.8 V	0	0	Ignored
	1	Ignored	0

¹ See Section [2.5.2 Power Scheme](#).

3.3 ROM Messages Printing Control

During the boot process, the messages by the ROM code can be printed to:

- **(Default) UART0 and USB Serial/JTAG controller**
- USB Serial/JTAG controller
- UART0

EFUSE_UART_PRINT_CONTROL and GPIO60 control ROM messages printing to **UART0** as shown in Table 3-5 *UART0 ROM Message Printing Control*.

Table 3-5. UART0 ROM Message Printing Control

UART0 ROM Code Printing	EFUSE_UART_PRINT_CONTROL	GPIO60
Enabled	0	Ignored
	1	0
	2	1
Disabled	1	1
	2	0
	3	Ignored

¹ **Bold** marks the default value and configuration.

EFUSE_DIS_USB_SERIAL_JTAG_ROM_PRINT controls the printing to **USB Serial/JTAG controller** as shown in Table 3-6 *USB Serial/JTAG ROM Message Printing Control*.

Table 3-6. USB Serial/JTAG ROM Message Printing Control

USB Serial/JTAG ROM Code Printing	EFUSE_DIS_USB_SERIAL_JTAG_ROM_PRINT
Enabled	0
Disabled	1

¹ **Bold** marks the default value and configuration.

3.4 JTAG Signal Source Control

The strapping pin GPIO37 can be used to control the source of JTAG signals during the early boot process. This pin does not have any internal pull resistors and the strapping value must be controlled by the external circuit that cannot be in a high impedance state.

As Table 3-7 *JTAG Signal Source Control* shows, GPIO37 is used in combination with EFUSE_DIS_PAD_JTAG, EFUSE_DIS_USB_JTAG, and EFUSE_JTAG_SEL_ENABLE .

Table 3-7. JTAG Signal Source Control

JTAG Signal Source	EFUSE_DIS_PAD_JTAG	EFUSE_DIS_USB_JTAG	EFUSE_JTAG_SEL_ENABLE	GPIO37
USB Serial/JTAG Controller	0	0	0	Ignored
	0	0	1	1
	1	0	Ignored	Ignored
JTAG pins ²	0	0	1	0
	0	1	Ignored	Ignored
JTAG is disabled	1	1	Ignored	Ignored

¹ **Bold** marks the default value and configuration.

² JTAG pins refer to MTDI, MTCK, MTMS, and MTDO.

4 Functional Description

4.1 System

This section describes the core of the chip's operation, covering its microprocessor, memory organization, system components, and security features.

4.1.1 Microprocessor and Master

This subsection describes the core processing units within the chip and their capabilities.

4.1.1.1 High-Performance CPU

The ESP32-S31 HP subsystem comprises two 32-bit RISC-V processor cores, HP CPU0 and HP CPU1, with the following features:

Feature List

- Five-stage pipeline that supports clock frequency of up to 320 MHz
- [RV32IMAFIC ISA](#) (instruction set architecture)
- Zc extensions (Zcb, Zcmp, and Zcmt)
- Zb extensions
- Custom AI and DSP extension (XespV)
- Custom hardware loop instructions (XespLoop)
- Compliant with RISC-V Sv32 virtual memory scheme
- Compliant with RISC-V Core Local Interrupt (CLINT)
- Compliant with RISC-V Core-Local Interrupt Controller (CLIC)
- Branch predictor BHT, BTB, and RAS
- Up to four hardware breakpoints/watchpoints
- Up to 32 PMP regions and 16 PMA regions
- Machine, Supervisor and User privilege modes
- USB/JTAG for debugging
- Compliant with RISC-V debug specification v0.13
- Offline trace debug that is compliant with RISC-V Trace Specification v2.0

4.1.1.2 RISC-V Trace Encoder (TRACE)

The RISC-V Trace Encoder in the ESP32-S31 chip provides a way to capture detailed trace information from the HP CPU's execution, enabling deeper analysis and optimization of the system. It connects to the HP CPU's instruction trace interface and compresses the information into smaller packets, which are then stored in internal SRAM.

Feature List

- Compatible with Efficient Trace for RISC-V v2.0
- Delta address mode and full address mode
- A filter unit
- Notifying an instruction address via debug trigger or filter unit
- Support for the following sideband signals to control trace data flow:
 - Debugging trigger to start or end encoder
 - When the hart is halted, the encoder can report the last packet and then stop
 - When the hart is reset, the encoder can report the last packet and then stop
 - Stalling the hart when FIFO is almost full
- Arbitrary address range of the trace memory size
- Configurable synchronization modes:
 - Synchronization counter counts by packet
 - Synchronization counter counts by cycle
 - Synchronization counter can be disabled
- Trace lost status to indicate packet loss
- Automatic restart after packet loss
- Memory writing in the loop or non-loop mode
- Two interrupts:
 - Triggered when the packet size exceeds the configured memory space
 - Triggered when a packet is lost
- FIFO (128 × 8 bits) to buffer packets
- AHB burst transmission with configurable burst length

4.1.1.3 Low-Power CPU

ESP32-S31 integrates an LP 32-bit RISC-V single-core processor (LP CPU). The LP CPU can assist the HP CPU in normal working mode, or replace the HP CPU to execute tasks when the system is in sleep mode. The LP CPU and LP memory remain powered on in Deep-sleep mode. Hence, the developer can store a program for the LP CPU in the LP memory to access LP IO, LP peripherals, and real-time timers in Deep-sleep mode.

Feature List

- Two-stage pipeline that supports a clock frequency of up to 40 MHz
- [RV32IMAC ISA](#) (instruction set architecture)
- 18 vector interrupts

- Debug module compliant with RISC-V External Debug Support Version 0.13 with external debugger support over an industry-standard JTAG/USB port
- Hardware trigger compliant with RISC-V External Debug Support Version 0.13 with up to 2 breakpoints/watchpoints
- Core performance metric events
- Can wake up the HP CPUs or trigger an interrupt to them
- Access to HP memory and LP memory
- Access to the entire peripheral address space

4.1.2 System DMA

This subsection describes the system DMA.

4.1.2.1 GDMA Controller (GDMA-AHB, GDMA-AXI)

General Direct Memory Access (GDMA) is a feature that allows peripheral-to-memory, memory-to-peripheral, and memory-to-memory data transfer at high speed. The CPU is not involved in the GDMA transfer and therefore is more efficient with less workload.

ESP32-S31 has two types of general-purpose DMA controllers, namely GDMA-AHB and GDMA-AXI, to directly access the AHB bus or the AXI bus respectively. One GDMA-AHB is placed on both the HP and LP sides.

Feature List

- Architecture:
 - GDMA-AHB: AHB bus architecture
 - GDMA-AXI: AXI bus architecture, which gives the possibility to complete up to eight transactions out of order and up to eight outstanding transactions
- Programmable length of data to be transferred in bytes
- Access via any address and size
- Alignment:
 - GDMA-AHB:
 - * Descriptor address: 1-word aligned
 - * Data address and length:
 - Internal memory and non-encrypted external memory address space: no requirements
 - Encrypted external memory address space: 16-byte aligned
 - GDMA-AXI:
 - * Descriptor address: 2-word aligned
 - * Data address and length:
 - Internal memory and non-encrypted external memory address space: no requirements

- Encrypted external memory address space: 16-byte aligned
- Linked list of descriptors
- INCR4/INCR8/INCR16 burst transfers when accessing memory via GDMA-AHB
- The GDMA-AHB on the HP side has 5 transmit channels and 5 receive channels; the GDMA-AHB on the LP side has 2 transmit channels and 2 receive channels
- GDMA-AXI has 3 transmit channels and 3 receive channels
- Software-configurable selection of peripheral requesting its service
- Configurable channel priority and weight arbitration
- Support for memory transfer
- Linked list switch interrupt mechanism (only supported by GDMA-AXI)

4.1.2.2 2D-DMA Controller (2D-DMA)

The 2D-DMA controller is a DMA (Direct Memory Access) dedicated to two-dimensional image processing. In addition to all the features of GDMA-AXI, it includes support for macroblock reordering and color space conversion (CSC) to better meet the data transfer requirements from JPEG and PPA. Notably, the 2D-DMA facilitates memory-to-memory transfers, enabling the movement of macroblocks between different segments of memory address space while concurrently performing color space conversion.

Feature List

- One AXI master interface
- Data transfer with unaligned starting addresses
- Memory-to-memory, peripheral-to-memory (RX), and memory-to-peripheral (TX) data transfer
- Four memory-to-peripheral channels, and three peripheral-to-memory channels
- Support for PPA and JPEG Codec
- Macroblock reordering
- Color space conversion
- Configurable channel priority and weight

4.1.3 Memory Organization

This subsection describes the memory arrangement to explain how data is stored, accessed, and managed for efficient operation.

Figure 4-1 illustrates the address mapping structure of ESP32-S31.

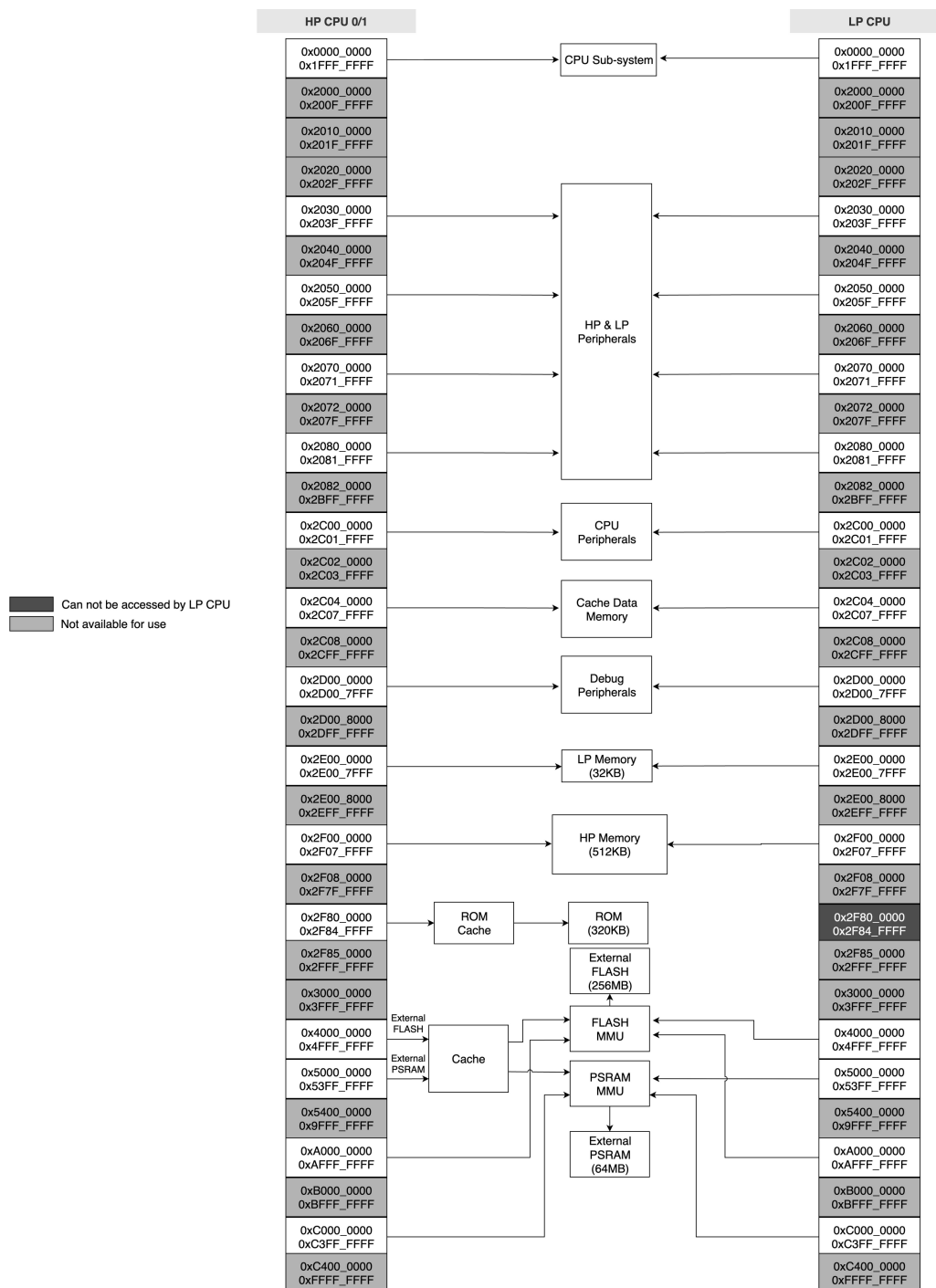


Figure 4-1. Address Mapping Structure

4.1.3.1 Internal Memory

The internal memory of ESP32-S31 refers to the memory integrated on the chip die or in the chip package, including ROM, SRAM, eFuse, and flash.

Feature List

- 320 KB of ROM for booting and core functions (the CPU accesses ROM indirectly through RomCache, not directly)

- 512 KB of SRAM for data and instructions
- 32 KB of low-power SRAM (LP SRAM) that can be accessed by the HP CPUs or LP CPU. It can retain data in Deep-sleep mode
- 4 Kbit of eFuse memory, with 1536 bits available for users.

4.1.3.2 External Memory

ESP32-S31 supports SPI, Dual SPI, Quad SPI, and QPI interfaces that allow connection to the external flash and PSRAM.

CPU's instruction memory space and read-only data memory space can map into the external flash and PSRAM of ESP32-S31. External flash supports up to 256 MB, while PSRAM supports up to 64 MB. ESP32-S31 supports hardware encryption/decryption based on XTS-AES to protect developers' programs and data in the external flash and PSRAM.

Feature List

- The 256 MB flash instruction space is mapped to external flash in 256 KB blocks, while the 64 MB PSRAM instruction space is mapped to external PSRAM in 64 KB blocks. Both support 32-bit instruction fetch.
- The 256 MB flash data space is mapped to external flash in 256 KB blocks, while the 64 MB PSRAM data space is mapped to external PSRAM in 64 KB blocks. External flash supports 8-bit, 16-bit, and 32-bit reads, while PSRAM supports 8-bit, 16-bit, and 32-bit read and write operations.

Note:

After ESP32-S31 is initialized, software can customize the mapping of external flash and PSRAM into the CPU address space.

4.1.3.3 eFuse Controller (eFuse)

The eFuse memory is a one-time programmable memory that stores parameters and user data, and the eFuse controller of ESP32-S31 is used to program and read this eFuse memory.

Feature List

- 4 Kbits in total, with 1536 bits reserved for users, e.g., encryption key and device ID
- One-time programmable storage
- Configurable write protection
- Configurable read protection
- Various hardware encoding schemes to protect against data corruption

4.1.3.4 Cache

The ESP32-S31 adopts a cache architecture in which each CPU (Core0 and Core1) has a private instruction cache, while both CPUs share a unified data cache.

Feature List

- 32 KB L1 instruction cache, 64 B block size, two-way set associative
- 64 KB L1 data cache, 64 B block size, two-way set associative, supporting write-through and write-back policies
- Cacheable and non-cacheable access
- Pre-load function
- Lock function
- Critical word first and early restart

4.1.4 System Components

This subsection describes the essential components that contribute to the overall functionality and control of the system.

4.1.4.1 GPIO Matrix and IO MUX

The ESP32-S31 chip features 60 GPIO pins, including 8 low-power (LP) GPIO pins and 52 high-performance (HP) GPIO pins. Each pin can be used as a general-purpose I/O, or be connected to an internal peripheral signal.

- Through HP GPIO matrix and HP IO MUX, HP peripheral input signals can be from any GPIO pins, and HP peripheral output signals can be routed to any GPIO pins.
- Through LP GPIO matrix and LP IO MUX, LP peripheral input signals can be from any LP GPIO pins, and LP peripheral output signals can be routed to any LP GPIO pins.

Together these modules provide highly configurable I/O. The 60 GPIO pins are numbered as follows: GPIO0 GPIO28, GPIO30 GPIO40, GPIO42 GPIO61.

- LP GPIO pins (GPIO0–GPIO7) can be used by either HP or LP peripherals.
- HP GPIO pins (GPIO8–GPIO28, GPIO30–GPIO40, GPIO42–GPIO61) can be used only by HP peripherals.

Feature List

HP GPIO matrix has the following features:

- A full-switching matrix between HP peripheral input/output signals and the GPIO pins
- 243 HP peripheral input signals sourced from the input of any GPIO pins
- 247 HP peripheral output signals routed to the output of any GPIO pins
- Signal synchronization for HP peripheral inputs based on HP IO MUX operating clock
- GPIO Filter hardware for input signal filtering
- Glitch Filter hardware for second-time filtering on input signal
- Sigma delta modulated (SDM) output
- GPIO simple input and output

- HP GPIO Wakeup

HP IO MUX has the following features:

- Control of 60 GPIOs (GPIO0–GPIO28, GPIO30–GPIO40, GPIO42–GPIO61) for HP peripherals.
- A configuration register provided for each GPIO pin, to control the pin's input/output, pull-up/pull-down, drive strength, and function selection.
- Better high-frequency digital performance achieved by routing some digital signals (SPI, EMAC) directly from HP IO MUX to peripherals.

LP GPIO matrix has the following features:

- A full-switching matrix between the LP peripheral input/output signals and the LP GPIO pins
- 9 LP peripheral input signals sourced from the input of any LP GPIO pins
- 26 LP peripheral output signals routed to the output of any LP GPIO pins
- GPIO Filter hardware for input signal filtering
- GPIO simple input and output
- LP GPIO Wakeup

LP IO MUX has the following features:

- Control of 8 LP GPIO pins (GPIO0–GPIO7) for LP peripherals.
- A configuration register provided for each LP GPIO pin, to control the pin's input/output, pull-up/pull-down, drive strength, function selection, and IO MUX selection.

4.1.4.2 Reset

ESP32-S31 provides four types of reset that occur at different levels, namely CPU Reset, Core Reset, System Reset, and Chip Reset. All reset types mentioned above (except Chip Reset) preserve the data stored in internal memory.

- Four reset types:
 - CPU Reset: resets CPU core. HP CPU0, HP CPU1, and LP CPU can be reset independently:
 - * HP CPU0 will be automatically released from reset after chip power-up.
 - * HP CPU1 is at reset by default after chip power-up, and needs to be manually released from reset.
 - * LP CPU is at reset after chip power-up, and needs to be manually released from reset by configuring the power management unit (PMU).
 - Core Reset: resets the whole digital system except for LP AON. HP core and LP core can be reset independently: HP Core Reset resets HP CPU0, HP CPU1, HP peripherals, HP GPIO, etc., and LP Core Reset resets LP CPU and LP peripherals.
 - System Reset: resets the whole digital system, including the LP system.
 - Chip Reset: resets the whole chip.
- Software reset and hardware reset:

- Software Reset: triggered via software by configuring the corresponding registers of CPU.
- Hardware Reset: triggered directly by the hardware.

4.1.4.3 Clock

ESP32-S31 clocks are mainly sourced from oscillator (OSC, including Resistor-Capacitor circuit), crystal (XTAL), and PLL circuit, and then processed by the dividers or selectors, which allows most functional modules to select their working clock according to their power consumption and performance requirements.

ESP32-S31 clock sources can be categorized based on frequency as follows:

- High-performance clocks, primarily used to provide operating clocks for HP CPU0/1 and HP digital peripherals:
 - BBPLL_CLK: 480 MHz internal PLL clock (reference clock: XTAL_CLK)
 - CPPLL_CLK: 320 MHz internal PLL clock (reference clock: XTAL_CLK)
 - MPLL_CLK: 500 MHz internal PLL clock (reference clock: XTAL_CLK)
 - APLL_CLK: 120 MHz internal PLL clock (reference clock: XTAL_CLK)
 - XTALX2_CLK: 80 MHz clock (reference clock: XTAL_CLK); used to provide a relatively high-speed clock for the HP domain when PLLs are not enabled
- Low-power clocks, primarily used for the low-power system and peripherals operating in low-power modes:
 - XTAL32K_CLK: 32 kHz external crystal oscillator clock
 - RC_SLOW_CLK: internal slow RC oscillator with adjustable frequency (typically 150 kHz)
 - EXT32K_CLK: external clock from a PAD, typically 32 kHz with limited accuracy
 - XTAL_CLK: 40 MHz external crystal oscillator clock
 - RC_FAST_CLK: internal fast RC oscillator with adjustable frequency (default 17.48 MHz)

4.1.4.4 Interrupt Matrix

ESP32-S31 interrupt matrix is used to route interrupt requests generated by peripherals and events to CPU interrupts.

Feature List

- Accepts 155 external interrupt sources as inputs
- Generates 32 external interrupts for the CPU as outputs
- Supports querying the current status of external interrupt sources
- Supports mapping multiple interrupt sources to a single CPU interrupt (i.e., shared interrupts)
- Supports interrupt delegation (lower-level interrupts can be remapped to higher privilege modes for CPU handling):
 - Supports User Mode interrupts being remapped to Supervisor Mode

- Supports User Mode interrupts being remapped to Machine Mode
- Supports Supervisor Mode interrupts being remapped to Machine Mode
- Supports error detection for interrupt delegation configuration

4.1.4.5 Event Task Matrix

The Event Task Matrix (ETM) peripheral contains 50 configurable channels. Each channel can map an event of any specified peripheral to a task of any specified peripheral. In this way, peripherals can be triggered to execute specified tasks without CPU intervention.

Feature List

- Receive various events from multiple peripherals
- Generate various tasks for multiple peripherals
- 50 independently configurable ETM channels
- An ETM channel can be set up to receive any event, and map it to any task
- Each ETM channel can be enabled independently. If not enabled, the channel will not respond to the configured event and generate the task mapped to that event
- Support for checking event and task status
- Peripherals supporting ETM include GPIO, LED PWM, general-purpose timers, RTC Timer, system timer, MCPWM, temperature sensor, ADC, I2S, LP CPU, GDMA-AHB, GDMA-AXI, 2D-DMA, and PMU

4.1.4.6 Power Management Unit

The ESP32-S31 has an advanced Power Management Unit (PMU). It can be flexibly configured to power up different power domains of the chip to achieve the best balance between chip performance, power consumption, and wakeup latency.

The integrated LP CPU allows the ESP32-S31 to operate in Deep-sleep mode with most of the power domains turned off, thus achieving extremely low-power consumption.

Configuring the PMU is a complex procedure. To simplify power management for typical scenarios, there are the following **predefined power modes** that power up different combinations of power domains:

- **Active mode** – The HP CPUs, RF circuits, and all peripherals are on. The chip can process data, receive, transmit, and listen.
- **Modem-sleep mode** – The HP CPUs are on, but the clock frequency can be reduced. The wireless connections can be configured to remain active as RF circuits are periodically switched on when required.
- **Light-sleep mode** – The HP CPUs stop running, and can be optionally powered on. The LP peripherals, as well as the LP CPU can be woken up periodically by the timer. The chip can be woken up via all wake up mechanisms: MAC, RTC timer, or external interrupts. Wireless connections can remain active. Some groups of digital peripherals can be optionally powered off.
- **Deep-sleep mode** – Only the LP system is powered on. Wireless connection data is stored in LP memory.

4.1.4.7 System Timer

The System Timer (SYSTIMER) in the ESP32-S31 chip is a 52-bit timer that can be used to generate tick interrupts for the operating system or as a general timer to generate periodic or one-time interrupts.

Feature List

- Two 52-bit counters and three 52-bit comparators
- Counters with an average clock frequency of 16 MHz
- Three types of independent interrupts generated according to alarm value
- Two alarm modes: target mode and period mode
- 52-bit alarm values and 26-bit alarm periods
- Automatic reload of counter value
- Counters can be stalled if the CPU is stalled or in on-chip-debugging mode
- Real-time alarm events

4.1.4.8 Timer Group (TIMG)

The Timer Group (TIMG) in the ESP32-S31 chip can be used to precisely time an interval, trigger an interrupt after a particular interval (periodically and aperiodically), or act as a hardware clock. ESP32-S31 has two timer groups, TIMGO and TIMG1, each consisting of one general-purpose timer and one Main System Watchdog Timer.

Feature List

- 16-bit prescaler
- 54-bit time-base counter programmable to incrementing or decrementing
- Able to read real-time value of the time-base counter
- Halt and resume the time-base counter
- Programmable alarm generation
- Timer value reload (auto-reload at an alarm or a software-controlled instant reload)
- Frequency calculation of slow clock for TIMGO
- Level interrupt generation
- Real-time alarm events
- Support several ETM tasks and events

4.1.4.9 Watchdog Timers (WDT)

ESP32-S31 contains three digital watchdog timers: one in each of the two timer groups (called Main System Watchdog Timers, or MWDT) and one in the LP system (called the RTC Watchdog Timer, or RWDT).

In SPI Boot mode, RWDT and the MWDT in timer group 0 are enabled automatically in order to detect errors that may occur during the flash boot process and facilitate recovery.

ESP32-S31 also has one analog watchdog timer: Super watchdog (SWD). It is an ultra-low-power circuit in analog domain that helps to prevent the system from operating in a sub-optimal state and resets the system if required.

Feature List

- Four stages, each with a separately programmable timeout value and timeout action
- Timeout actions:
 - MWDT: interrupt, HP CPU reset, HP core reset
 - RWDT: interrupt, HP CPU reset, HP core reset, system reset
- Flash boot protection under SPI Boot mode at stage 0:
 - MWDT0: HP core reset upon timeout
 - RWDT: system reset upon timeout
- Write protection that makes WDT register read only unless unlocked
- 32-bit timeout counter
- Clock source:
 - MWDT: PLL_F80M_CLK, RC_FAST_CLK or XTAL_CLK
 - RWDT: LP_DYN_SLOW_CLK

4.1.4.10 RTC Timer

RTC Timer is an important module for implementing low power management of ESP32-S31. Based on a 48-bit readable counter, RTC Timer is mainly used as a system timer in low power mode when the timer peripheral in the HP system is unavailable. It also allows for configuring timer interrupts and logging the time when specific events happen in the system.

Feature List

- 48-bit counter
- Time logging when one of the following events happens:
 - HP system reset
 - CPU enters stall state
 - CPU exits stall state
 - Crystal powers up
 - Crystal powers down
- Time logging through register configuration
- Occurrence time cached of the most recent two specific events

- Generation of interrupts at target times, which are configurable. It is also possible to configure two target times simultaneously.
- Uninterrupted operation during any reset or sleep mode, except for power-on reset of LP system.
- Supports counter ticks that trigger the system timer to lock time.

4.1.4.11 Permission Control (PMS)

The Permission Control module in ESP32-S31 is responsible for managing access permissions to memory and peripheral registers. It consists of two parts: PMP (Physical Memory Protection) and APM (Access Permission Management).

Feature List

- access permission management for ROM, HP memory, HP peripheral, and LP peripheral address spaces
- APM supports each master (such as DMA) to select one of the four security modes
- access permission configuration for up to 32 address ranges
- individual permission configuration for each register
- Interrupt function and exception information record

4.1.4.12 System Registers

The System Registers in the ESP32-S31 chip are used to configure various auxiliary chip features.

Feature List

- Controls external memory encryption
- Controls HP/LP core debugging
- Controls bus timeout protection
- Software interrupts
- Memory power management
- Clock control
- PAD BIST control

4.1.4.13 Debug Assistant

The Debug Assistant provides a set of functions to help locate bugs and issues during software debugging. It offers various monitoring capabilities and logging features to assist in identifying and resolving software errors efficiently.

Feature List

- **Region read/write monitoring:** Monitors whether the high-performance dual-core processors (HP CPU0 and HP CPU1) read from or write to a specified memory address space on the bus. A detected read or write in the monitored address space triggers an interrupt.

- **Stack pointer (SP) monitoring:** Monitors whether the SP exceeds the specified address space. A bounds violation triggers an interrupt.
- **Program counter (PC) logging:** Records the PC value. The last PC value at the most recent reset of the HP CPU n (Below, $n = 0, 1$) can be retrieved.
- **Lockup monitoring:** Monitors whether HP CPU n encounters a lockup, and records the cause of the first exception, the trap value (tval), instruction PC (iaddr), and current privilege level (priv) when the first abnormal event occurs.
- **Bus access logging:** Records information about bus access. When the HP CPU n or the Direct Memory Access controller (DMA) writes a specified value, the Debug Assistant module records the data type, address of the write operation, and the PC value when the write is performed by the HP CPU n , then stores this information in the HP SRAM.

4.1.4.14 LP Mailbox

ESP32-S31 integrates an LP Mailbox module that provides an efficient inter-core communication mechanism between the LP CPU and HP CPU0/1. The LP Mailbox module comprises sixteen 32-bit message registers that the LP CPU and HP CPU0/1 can use to store and exchange messages. Inter-core communication between LP CPU and HP CPU0/1 is achieved through an interrupt mechanism implemented in the LP Mailbox module.

Feature List

- Sixteen 32-bit message registers for inter-core communication
- LP CPU external interrupt signal
- HP CPU0/1 external interrupt signal

4.1.4.15 Brown-out Detector

With the Brown-out detector, ESP32-S31 monitors the voltage levels of pin VDDPST. If the voltage on these pins drops below the predefined threshold (defaulting to 2.4 V), the detector triggers signals to shut down certain power-consuming blocks (e.g., flash), ensuring that the digital module has sufficient time to save and transfer important data.

Feature List

- Monitors the voltage level of pin VDDPST
- Two configurable monitoring modes
 - Mode 0: The brown-out detector triggers interrupts when the brown-out counter reaches the predefined threshold and selects the reset mode according to the configuration.
 - Mode 1: The brown-out detector triggers a system reset when the voltage falls below the threshold.
- Configurable voltage-monitoring thresholds and noise tolerance

4.1.5 Cryptography and Security Component

This subsection describes the security features incorporated into the chip, which safeguard data and operations.

4.1.5.1 AES Accelerator (AES)

ESP32-S31 integrates a hardware AES (Advanced Encryption Standard) accelerator that performs data encryption and decryption using the AES algorithm. It supports two working modes: typical AES and DMA-AES. Overall, compared with software-based AES computation, the hardware AES accelerator significantly improves processing speed. In addition, the ESP32-S31 AES accelerator includes a configurable anti-side-channel attack (anti-DPA) feature, providing enhanced security.

Feature List

- Typical AES working mode
 - AES-128/AES-256 encryption and decryption operations compliant with [NIST FIPS 197](#)
- DMA-AES working mode
 - AES-128/AES-256 encryption and decryption operations compliant with [NIST FIPS 197](#)
 - Block cipher mode compliant with [NIST SP 800-38A](#)
 - * ECB (Electronic Codebook)
 - * CBC (Cipher Block Chaining)
 - * OFB (Output Feedback)
 - * CTR (Counter)
 - * CFB8 (8-bit Cipher Feedback)
 - * CFB128 (128-bit Cipher Feedback)
 - GCM (Galois/Counter Mode)
 - Interrupt on completion of computation
- Configurable anti-side-channel attack (anti-DPA) capability

4.1.5.2 ECC Accelerator (ECC)

Elliptic Curve Cryptography (ECC) is an approach to public-key cryptography based on the algebraic structure of elliptic curves. ECC allows smaller keys compared to RSA cryptography while providing equivalent security.

ESP32-S31's ECC Accelerator can complete various calculations based on different elliptic curves, thus accelerating the ECC algorithm and ECC-derived algorithms (such as ECDSA).

Feature List

- Four different elliptic curves, namely, P-192, P-256, and P-384 defined in [FIPS 186-5](#) and SM2 defined in [GM/T 0003.2-2012](#)

- 11 working modes
- Enhanced anti-attack performance
- Interrupt upon completion of calculation and interrupt control

4.1.5.3 HMAC Accelerator (HMAC)

The Hash-based Message Authentication Code (HMAC) module computes Message Authentication Codes (MACs) using hash algorithm SHA-256 and keys as described in RFC 2104. The 256-bit HMAC key is stored in an eFuse key block and can be set as read-protected; that is, the key is not accessible from outside the HMAC accelerator.

Feature List

- Standard HMAC-SHA-256 algorithm
- HMAC-SHA-256 calculation based on key in eFuse,
 - whose result cannot be accessed by software in downstream mode for high security
 - whose result can be accessed by software in upstream mode
- Generates required keys for the RSA Digital Signature Peripheral (RSA_DS) in downstream mode
- Re-enables soft-disabled JTAG in downstream mode

4.1.5.4 RSA Digital Signature Peripheral (RSA_DS)

The RSA Digital Signature Peripheral (RSA_DS) verifies the authenticity and integrity of a message using a cryptographic algorithm. This can be used to, for example, validate a device's identity to a server.

ESP32-S31 includes an RSA Digital Signature Peripheral (RSA_DS) providing hardware acceleration for RSA-based message signing operation. The HMAC peripheral is used as the key derivation function (KDF) to output *RSA_DS_KEY* using either a key stored in eFuse or deployed by the Key Manager as the input key. Subsequently, the RSA_DS peripheral uses *RSA_DS_KEY* to decrypt the pre-encrypted parameters and calculate the signature. The whole process happens in hardware so that neither the decryption key for the RSA parameters nor the input key for the HMAC key derivation function can be seen by users during signature calculation.

Feature List

- Hardware RSA signature calculation as $Z = X^Y \bmod M$, with private-key operands up to 4096 bits
- Private key parameters are stored encrypted in flash, and decrypted inside the RSA_DS peripheral using *RSA_DS_KEY*; plaintext private key material is not visible to software
- SHA-256 digest over private key data and related parameters for tamper detection
- Hardware-provisioned *RSA_DS_KEY* (not readable by software), from HMAC-SHA256 using an eFuse-stored *HMAC_KEY* or from the Key Manager

4.1.5.5 SHA Accelerator (SHA)

ESP32-S31 integrates an SHA accelerator, which is a hardware device that speeds up the SHA algorithm significantly, compared with an SHA algorithm implemented solely in software. The SHA accelerator integrated in ESP32-S31 has two working modes, Typical SHA and DMA-SHA.

Feature List

- The following hash algorithms introduced in [FIPS PUB 180-4 Spec.](#)
 - SHA-1
 - SHA-224
 - SHA-256
 - SHA-384
 - SHA-512
 - SHA-512/224
 - SHA-512/256
 - SHA-512/t
- Supports [SM3 cryptographic hash algorithm](#)
- Two working modes
 - Typical SHA
 - DMA-SHA
- Interleaved function when working in Typical SHA working mode
- Interrupt function when working in DMA-SHA working mode

4.1.5.6 RSA Digital Signature Peripheral (RSA_DS)

Digital signature technology uses cryptographic algorithms to verify the authenticity and integrity of messages. It can also be used to authenticate a device to a server or to verify whether a message has been tampered with.

ESP32-S31 includes an RSA Digital Signature Peripheral (RSA_DS) that provides hardware acceleration for efficiently generating RSA-based digital signatures. The RSA_DS peripheral uses the RSA_DS_KEY (generated by HMAC or provisioned by the key manager) to decrypt pre-encrypted parameters and compute the signature. All these operations occur entirely in hardware. During the process, sensitive data such as the key for decrypting RSA parameters, the input/output keys of the HMAC key derivation function, or keys provisioned by the key manager remain inaccessible to the user.

Feature List

- RSA digital signatures with key length up to 4096 bits
- Encrypted private key data, only decryptable by the RSA_DS peripheral
- SHA-256 digest to protect private key data against tampering by an attacker

4.1.5.7 ECDSA Digital Signature Peripheral (ECDSA_DS)

In cryptography, the Elliptic Curve Digital Signature Algorithm (ECDSA) offers a variant of the Digital Signature Algorithm (DSA) which uses elliptic-curve cryptography.

ESP32-S31's ECDSA Digital Signature Peripheral (ECDSA_DS) provides a secure and efficient environment for computing ECDSA signatures. It offers fast computations while ensuring the confidentiality of the signing process to prevent information leakage. This makes it a valuable tool for applications that require high-speed cryptographic operations with strong security guarantees. By using ECDSA_DS, users can be confident that their data is protected without sacrificing performance.

Feature List

- Digital signature generation and verification
- Four elliptic curves, namely P-192, P-256, and P-384 as defined in [FIPS 186-5](#), and SM2 as defined in [GM/T 0003.2-2012](#)
- Multiple message hash algorithms for ECDSA_DS operations, including SHA-224, SHA-256, SHA-384, SHA-512, SHA-512-224, and SHA-512-256 as defined in [FIPS PUB 180-4 Spec](#), and SM3 as defined in [GM/T 0004-2012](#)
- State-dependent register access control in different operation states to ensure information security

4.1.5.8 Secure Debug Controller (SDC)

The Secure Debug Controller (SDC) module implements a hardware-enforced secure debug unlocking protocol for deployed chips. SDC provides the only authorized path to selectively re-enable debug and download interfaces, ensuring chips can enter debug mode only after being returned to authorized service centers. This capability is enabled by eFuse configuration, preventing software-only bypass.

SDC unlocking uses the on-chip SHA and ECDSA_DS peripherals for cryptographic acceleration. The unlock process follows a two-phase challenge-response protocol: in the request phase, SDC calculates a chip-unique CHIP_INFO to generate the certificate; in the verification phase, the certificate is loaded into SDC for hash and signature verification. Any verification failure puts SDC into a terminal lock state, and further attempts require a power cycle.

Feature List

- Three independently selectable re-enable outputs: JTAG re-enable, Download mode, and Force SPI boot
- Two operating modes: Certificate Request and Certificate Verification
- Optional nonce integration for per-session challenge uniqueness
- Hash mode: SHA-256
- Public-key signature mode: ECDSA P-256

4.1.5.9 External Memory Encryption and Decryption (XTS_AES)

The ESP32-S31 integrates an External Memory Encryption and Decryption module that complies with the XTS-AES standard algorithm specified in [IEEE Std 1619-2007](#), providing security for users' application code and

data stored in the external memory (flash and RAM). Users can store proprietary firmware and sensitive data (e.g., credentials for gaining access to a private network) in the external flash, or store general data in the external RAM.

Feature List

- General XTS-AES algorithm, compliant with [IEEE Std 1619-2007](#)
- Software-based manual encryption
- High-speed auto encryption without software's participation
- High-speed auto decryption without software's participation
- Encryption and decryption functions jointly enabled/disabled by register configuration, eFuse parameters, and boot mode
- Configurable anti-side-channel attack (anti-DPA) capability
- Independent keys for flash and PSRAM respectively

4.1.5.10 Random Number Generator (RNG)

The ESP32-S31 contains a true random number generator (TRNG), which generates 32-bit random numbers that can be used for cryptographic operations, among other things.

The TRNG in ESP32-S31 generates true random numbers, which means random numbers generated from a physical process, rather than by means of an algorithm. No number generated within the specified range is more or less likely to appear than any other number.

Feature List

- RNG entropy source
 - thermal noise from SAR ADC
 - an asynchronous clock mismatch
 - BUF_CHAIN

4.1.5.11 Key Manager

ESP32-S31 stores and deploys keys with the Key Manager as the security core. Key Manager uses the unique physically unclonable function (PUF) of each chip to generate the hardware unique key (HUK) which is unique to the chip and serves as the root of trust (RoT) for the chip. HUK is automatically generated each time the chip is powered on and disappears when the chip is powered off. In this way, Key Manager secures key storage and deployment.

Key Manager of ESP32-S31 stores key information (non-plaintext information for recovering the key) in external memory, realizing flexible key management functionalities such as unlimited key storage and dynamic key switching.

Feature List

The HUK Generator has the following features:

- HUK Generation Mode:
 - Generates a new HUK and its recovery information
- HUK Recovery Mode:
 - Recovers a deployed HUK with its recovery information
- Prompt for HUK recovery error
- Prompt for HUK risk level

The Key Manager has the following features:

- Unlimited number of keys
- Specified private key deployment (AES Deploy Mode):
 - Users specify the value of the key
- Negotiated private key deployment (ECDH0 Deploy Mode):
 - Highest security mode: there is no need to worry about the leaks of data in external channels
 - Requires chip initiation to obtain the private key
 - Negotiates the key value between each chip and the user
- Negotiated private key deployment (ECDH1 Deploy Mode):
 - Provides auxiliary key software/scripts to users
 - No need to boot the chip to obtain the private key
- Random key deployment (Random Deploy Mode):
 - Deploys a hardware-generated random key with nobody knowing the exact value
- Private key recovery deployment (Private Key Recovery Mode):
 - Recovers exactly the same key by entering the key information generated during deployment
- Key information export (*key_info* Export Mode):
 - Generates unique key information for the same key each time

4.1.5.12 Power Glitch Detector

ESP32-S31 can monitor the voltage of the power supply in real time. When a voltage glitch occurs, the chip will reset immediately to prevent power glitch attacks.

Feature List

- Detect four supply voltages, including VDDPST and VDDA
- Configurable threshold for power glitch (around 2.4 V by default)
- Can be enabled through eFuse at power-up

4.1.5.13 Secure Boot

The Secure boot feature in the ESP32-S31 chip ensures that only the signed firmware can be booted.

Feature List

- Supported signature types
 - ECDSA P-192 signature
 - ECDSA P-256 signature
 - ECDSA P-384 signature

For details, see ESP-IDF Programming Guide > [Secure Boot v2](#).

4.2 Peripherals

This section describes the chip's peripheral capabilities, covering connectivity interfaces and on-chip sensors that extend its functionality.

4.2.1 Image and Voice Processing

This subsection describes the peripherals for image and voice processing.

4.2.1.1 JPEG Codec

ESP32-S31's JPEG codec is an image codec, which is based on the JPEG baseline standard, for compressing (encoding) and decompressing (decoding) images to reduce the bandwidth required to transmit images or the space required to store images, making it possible to process large-resolution images.

Feature List

When used as an encoder, the JPEG codec has the following features:

- Integrated discrete cosine transform algorithm
- Integrated canonical Huffman coding
- RGB888, RGB565, YUV444, YUV422, YUV420 and GRAY as original input image formats
- Supports converting (if needed) and compressing RGB888, RGB565, or YUV444 images into YUV444, YUV422, or YUV420 formats, and supports converting (if needed) and compressing YUV422 images into YUV422 or YUV420 formats. Compression is only available for YUV444, YUV422, and YUV420 formats
- Four configurable quantization coefficient tables with 8-bit or 16-bit precision
- Performance:
 - Still image compression: up to 4K resolution
 - Dynamic image compression: up to 720P@30fps (excluding header encoding time)
- Automatically added stuffed zero byte
- Automatically added EOI marker

When used as a decoder, the JPEG codec has the following features:

- Integrated inverse discrete cosine transform algorithm
- Integrated Huffman decoding
- Supported image formats for compressed bitstream decoding: YUV444, YUV422, YUV420, and GRAY.
- Four configurable quantization coefficient tables with 8-bit or 16-bit precision
- Two DC and two AC Huffman tables
- Supports image decoding of any resolution. However, the resolution of the output decoded image differs from the format of the input image:
 - YUV444, GRAY: both the horizontal and vertical resolutions of the output decoded image are multiples of 8, i.e., 150 × 150 images with an output resolution of 152 × 152

- YUV422: the horizontal resolution of the output decoded image is the multiples of 16 and the vertical resolution is multiples of 8, i.e., 150 × 150 images with an output resolution of 160 × 152
- YUV420: both the horizontal and vertical resolutions of the output decoded image are multiples of 16, i.e., 150 × 150 images with an output resolution of 160 × 160
- Performance:
 - Still image decoding: up to 4K resolution
 - Dynamic image decoding: up to 720P@30fps (excluding header parsing time)

Pin Assignment

The JPEG Codec does not interact directly with IOs, so it has no pins assigned.

4.2.1.2 Pixel-Processing Accelerator (PPA)

ESP32-S31 includes a pixel-processing accelerator (PPA) with scaling-rotation-mirror (SRM) and image blending (BLEND) functionalities.

Feature List

- Image rotation, scaling, and mirroring by SRM:
 - Input formats: ARGB8888, RGB888, RGB565, YUV422, YUV420, GRAY
 - Output formats: ARGB8888, RGB888, RGB565, YUV422, YUV420, GRAY
 - Counterclockwise rotation angles: 0°, 90°, 180°, 270°
 - Horizontal and vertical scaling with scaling factors of 4-bit integer part and 8-bit fractional part
 - Horizontal and vertical mirroring
- Blending two layers of the same size and filling images with specific pixels by BLEND:
 - Foreground input formats: ARGB8888, RGB888, RGB565, L4, L8, A4, A8
 - Background input formats: ARGB8888, RGB888, RGB565, YUV422, YUV420, GRAY, L4, L8
 - Output formats: ARGB8888, RGB888, RGB565, YUV422, YUV420, GRAY
 - Layer blending based on the Alpha channel. If layers lack an Alpha channel, it can be provided through register configuration.
 - Special color filtering by setting color-key ranges of foreground and background layers

Pin Assignment

The pixel-processing accelerator does not directly interact with IOs, so it has no pins assigned.

4.2.1.3 Audio Sample Rate Converter (ASRC)

Audio Sample Rate Converter (ASRC) is an accelerator-type audio processing module used for high-quality conversion of audio signals between different sample rates. It enables collaboration between devices with

varying sample rate standards. This module typically interacts with on-chip or off-chip memory through a DMA interface to achieve efficient transmission and processing of audio data.

Feature List

- Supports conversion between 8 kHz, 16 kHz, 32 kHz, 44.1 kHz, and 48 kHz
- Input and output signal width is 16 bits, using Q1.15 format
- Sample rate conversion uses two integer resamplers and one fractional sample rate converter (FRC) in a cascade, with the cascade configurable via software:
 - The integer resampler's conversion factor can be set to 2 or $\frac{1}{2}$
 - The FRC supports conversion factors configurable between 0 and 2 (excluding 0 and 2)
- Supports multiple channel modes, including:
 - Mono receive (Rx) / Mono transmit (Tx)
 - Stereo receive (Rx) / Stereo transmit (Tx), processed in parallel
 - Mono receive / Stereo transmit
 - Stereo receive / Mono transmit
- Supports input data of any length, and generates EOF (End of Frame) flags based on output length
- When two channels are used independently, each channel supports the following modes:
 - Mono input, mono output
 - Mono input, stereo output (both channels use the same data)
 - Stereo input, mono output (choose one channel for processing)
- When both channels are used together, each channel can be configured as stereo input or output

4.2.1.4 CORDIC Accelerator (CORDIC)

ESP32-S31 includes a hardware CORDIC accelerator. Its core principle is to iteratively approach a target angle by executing a sequence of fixed-angle rotations associated with computation cycles, enabling mathematical operations such as trigonometric function calculations.

Feature List

- Supports q1.15 and q1.31 fixed-point formats
- Supports circular, linear, and hyperbolic systems
- Supports rotation mode and vectoring mode
- Supported functions: sin, cos, sinh, cosh, atan(x), atan(y/x), atanh, modulus, square root, and natural logarithm
- Configurable computation precision (number of operation cycles)
- Supports polling-based result readout
- Supports interrupt-based result readout

- Supports direct DMA connection mode

Pin Assignment

The CORDIC accelerator does not need direct interaction with IO and therefore requires no dedicated pin assignment.

4.2.1.5 LCD and Camera Controller (LCD_CAM)

The LCD and Camera controller (LCD_CAM) on the ESP32-S31, consisting of an independent LCD control module and a camera control module, is a versatile component designed to facilitate interfacing with both LCDs and cameras.

Feature List

- Operation modes:
 - LCD master TX mode
 - Camera slave RX mode
 - Camera master RX mode
- Simultaneous connection to an external LCD and a camera
- External LCD interface:
 - 8/16/24-bit parallel output modes
 - RGB, MOTO6800, and I8080 LCD formats
 - LCD data retrieved from internal memory or external memory via GDMA
- External camera (DVP image sensor) interface:
 - 8/16-bit parallel input modes
 - Camera data stored in internal or external memory via GDMA
- Interrupt support

Pin Assignment

For CAM and LCD interfaces of the Camera-LCD controller, the pins used can be chosen from any GPIOs via the GPIO Matrix.

4.2.2 Connectivity Interface

This subsection describes the connectivity interfaces on the chip that enable communication and interaction with external devices and networks.

4.2.2.1 UART Controller (UART)

The UART controllers in ESP32-S31 are used for asynchronous serial data transmission and reception between the chip and external UART devices. ESP32-S31 includes four UART controllers in the main system and one low-power LP UART.

Feature List

- Programmable baud rates up to 5 MBaud
- RAM shared by TX FIFOs and RX FIFOs
- Support for various lengths of data bits and stop bits
- Parity bit support
- Special character AT_CMD detection
- RS485 protocol support (not applicable to LP UART)
- IrDA protocol support (not applicable to LP UART)
- High-speed data communication using GDMA (not applicable to LP UART)
- Receive timeout feature
- UART as the wake-up source
- Software and hardware flow control

Pin Assignment

UART0 and LP UART each have fixed direct-connected pins via the HP IO MUX and LP IO MUX. They also support mapping to other pins through the GPIO Matrix and LP GPIO Matrix. UART1, UART2, and UART3 are routed to any HP GPIO pins via the GPIO Matrix.

4.2.2.2 SPI Controller (SPI)

The Serial Peripheral Interface (SPI) is a synchronous serial interface commonly used for communicating with external peripherals. The ESP32-S31 chip integrates four SPI controllers:

- MSPI controller, including two sub-controllers
 - FLASH MSPI controller
 - * FLASH MSPI SPI0
 - * FLASH MSPI SPI1
 - PSRAM MSPI controller
 - * PSRAM MSPI SPI0
 - * PSRAM MSPI SPI1
- General Purpose SPI2 (GP-SPI2)
- General Purpose SPI3 (GP-SPI3)
- Low-Power SPI (LP-SPI)

Feature List

GP-SPI has the following features:

- Works as master or as slave

- Half- and full-duplex communications
- CPU- and DMA-controlled transfers
- Various data modes
 - **GP-SPI2**
 - * 1-bit SPI mode
 - * 2-bit Dual SPI mode
 - * 4-bit Quad SPI mode
 - * QPI mode
 - * 8-bit Octal SPI mode (available only when GP-SPI2 works as a master)
 - * OPI mode (available only when GP-SPI2 works as a master)
 - **GP-SPI3**
 - * 1-bit SPI mode
 - * 2-bit Dual SPI mode
 - * 4-bit Quad SPI mode
 - * QPI mode
- Configurable module clock frequency
 - Master: up to 80 MHz
 - Slave: up to 60 MHz
- Configurable data length
 - CPU-controlled transfer as master or as slave: 1–64 bytes
 - DMA-controlled single transfer as master: 1–32 KB
 - DMA-controlled configurable segmented transfer as master: data length is unlimited
 - DMA-controlled single transfer or segmented transfer as slave: data length is unlimited
- Configurable bit read/write order
- Independent interrupts for CPU-controlled transfer and DMA-controlled transfer
- Configurable clock polarity and phase
- Four SPI clock modes: mode 0–mode 3
- Multiple CS lines as master
 - **GP-SPI2**: CS0–CS5
 - **GP-SPI3**: CS0–CS2
- Able to communicate with SPI devices, such as a sensor, a screen controller, as well as a flash or RAM chip

LP-SPI is a simplified version of GP-SPI and has a subset of GP-SPI's features:

- Works as a master or as a slave

- Half- and full-duplex communications
- CPU-controlled transfer
- 1-bit SPI data mode
- Configurable module clock frequency:
 - Master: up to 40 MHz
 - Slave: up to 40 MHz
- Configurable data length:
 - CPU-controlled transfer as master or as slave: 1–64 bytes
- Configurable bit read/write order
- Interrupts for CPU-controlled transfer
- Configurable clock polarity and phase
- Four SPI clock modes: mode 0–mode 3
- One CS line as master: CS0
- Wake-up feature as slave (the only new feature compared with GP-SPI)

Pin Assignment

The Flash SPI interface uses the dedicated digital pins 36–42.

The GP-SPI2 controller includes one four-line interface and one eight-line interface. The pins connected to the four-line interface are multiplexed via the IO MUX with GPIO20 to GPIO25, or GPIO50 to GPIO52, as well as the JTAG interface. The pins connected to the eight-line interface are multiplexed with GPIO9 to GPIO19, which also serves as the first set of RMII interface pins for the EMAC. If high-speed performance is not critical for the GP-SPI2 interface, you can select pins from any GPIOs via the GPIO Matrix.

For GP-SPI3, the pins used can be chosen from any GPIOs via the GPIO Matrix.

The pins for the LP-SPI interface can be chosen from any pins via the LP GPIO Matrix.

4.2.2.3 I2C Controller (I2C)

ESP32-S31 has two HP_I2C bus interfaces and one LP_I2C bus interface. The HP_I2C interfaces can operate in either I2C master or slave mode, while the LP_I2C interface supports master mode only.

Feature List

- Standard mode (100 Kbit/s)
- Fast mode (400 Kbit/s)
- Up to 800 Kbit/s (constrained by SCL and SDA pull-up strength)
- 7-bit and 10-bit addressing mode
- Dual address mode
- 7-bit broadcast address

Pin Assignment

For I2C, the pins used can be chosen from any GPIOs via the GPIO Matrix.

4.2.2.4 I2S Controller (I2S)

The I2S Controller in the ESP32-S31 chip provides a flexible communication interface for streaming digital data in multimedia applications, particularly digital audio applications.

Feature List

- Master mode and slave mode
- Full-duplex and half-duplex communications
- Separate TX and RX units that can work independently or simultaneously
- A variety of audio standards supported:
 - TDM Philips standard
 - TDM MSB alignment standard
 - TDM PCM standard
 - PDM standard
- Various TX/RX modes
 - TDM TX mode, up to 16 channels supported
 - TDM RX mode, up to 16 channels supported
 - PDM TX mode
 - * Raw PDM data transmission
 - * PCM-to-PDM data format conversion, up to 2 channels supported
 - PDM RX mode
 - * Raw PDM data reception
- Configurable clock source with frequency up to 96 MHz
- Configurable high-precision sample clock with a variety of sampling frequencies supported
- 8/16/24/32-bit data width
- Synchronous counter in TX mode
- ETM feature
- Direct memory access
- Standard I2S interface interrupts

Pin Assignment

The pins for the I2S controller can be chosen from any GPIOs via the GPIO Matrix.

4.2.2.5 Pulse Count Controller (PCNT)

The Pulse Count controller (PCNT) in ESP32-S31 captures pulses and counts pulse edges in seven modes.

Feature List

- Four independent pulse counters (units) that count from 1 to 65535
- Each unit consists of two independent channels sharing one pulse counter
- All channels have input pulse signals (e.g., sig_ch0_un) with their corresponding control signals (e.g., ctrl_ch0_un)
- Independently filter glitches of input pulse signals (sig_ch0_un and sig_ch1_un) and control signals (ctrl_ch0_un and ctrl_ch1_un) on each unit
- Each channel has the following parameters:
 1. Selection between counting on positive or negative edges of the input pulse signal
 2. Configuration to Increment, Decrement, or Disable counter mode for control of signal's high and low states
 3. Step count alert triggered by setting the upcount/downcount step threshold
 4. Clearing of the pulse count controller value by setting the clear register or sending a clear signal through GPIO input
 5. Generation and recording of a corresponding event signal for each counter mode, with the ability to report it to the interrupt task
- Maximum frequency of pulses: $\frac{f_{APB_CLK}}{2}$

Pin Assignment

The pins for the Pulse Count controller can be chosen from any GPIOs via the GPIO Matrix.

4.2.2.6 USB 2.0 High-Speed OTG

The ESP32-S31 chip features a USB 2.0 High-Speed On-The-Go peripheral (OTG_HS) with an integrated transceiver. This OTG_HS complies with the USB 2.0 specification, OTG Revision 1.3, and OTG Revision 2.0 specifications. The interface supports USB 2.0 High-Speed mode (480 Mbit/s), Full-Speed mode (12 Mbit/s), and Low-Speed mode (1.5 Mbit/s).

- When OTG_HS operates in High-Speed or Full-Speed modes, it can be configured as either a Host or a Device.
- When OTG_HS operates in Low-Speed mode, it can only be configured as a Host.

Feature List

General Features

- USB 2.0 specification, OTG Revision 1.3 and OTG Revision 2.0 specifications

- High-Speed, Full-Speed, and Low-Speed data rates
- As a host and a device in High-Speed mode and Full-Speed mode
- Dynamic FIFO (DFIFO) sizing, each device EP/host channel can dynamically allocate a maximum of 4 KB FIFO.
- Up to 8 non-periodic transactions and 16 periodic transactions per microframe
- Multiple modes of memory access
 - Scatter/Gather DMA mode
 - Buffer DMA mode
 - Slave Mode
- Integrated UTMI High-Speed transceiver

Device Mode Features

- Endpoint 0 always present, bi-directional, consisting of EPO IN and EPO OUT
- 15 additional endpoints 1–15, configurable as IN or OUT
- Maximum of eight IN endpoints concurrently active at any time, including EPO IN
- All OUT endpoints share a single RX FIFO
- Each IN endpoint has a dedicated TX FIFO

Host Mode Features

- 16 host channels
- RX FIFO: shared by all periodic and non-periodic transactions
- Two TX FIFO:
 - One shared by all non-periodic transactions
 - One shared by all periodic transactions
- All of the above FIFOs share a 4 KB RAM.
- The size of each FIFO is configurable, with a maximum of 4 KB.

Pin Assignment

The pins connected to USB2 OTG PHY DM (USB_D-) and USB2 OTG PHY DP (USB_D+) signals of USB 2.0 High-Speed OTG are dedicated pin 44 and pin 45. Other signals can be routed to any GPIOs via the GPIO matrix.

4.2.2.7 USB Serial/JTAG Controller (USB_SERIAL_JTAG)

ESP32-S31 contains a USB Serial/JTAG controller. This unit can be used to program the SoC's flash, read program output, as well as attach a debugger to the running program. All of these are possible for any computer with a USB host without any active external components.

Feature List

- USB 2.0 full speed compliant, capable of up to 12 Mbit/s transfer speed (note that this controller does not support the faster 480 Mbit/s high-speed transfer mode)
- CDC-ACM virtual serial port and JTAG adapter functionality
- programming the chip's flash
- CPU debugging with compact JTAG instructions
- a full-speed USB PHY integrated in the chip

Pin Assignment

The pins for the USB Serial/JTAG controller are multiplexed with GPIO33 and GPIO34 via IO MUX.

4.2.2.8 Ethernet Media Access Controller (EMAC)

By using the external Ethernet PHY (physical layer), ESP32-S31 can send and receive data via Ethernet MAC (Media Access Controller) according to the IEEE 802.3 standard.

ESP32-S31 Ethernet MAC complies with the following standards:

- IEEE 802.3-2002 for Ethernet MAC
- IEEE 1588-2008 standard for precise networked clock synchronization
- IEEE 802.3 standard Media Independent Interface (MII), Reduced Media Independent Interface (RMII), and Reduced Gigabit Media Independent Interface (RGMII)
- IEEE 802.3az-2010 for Energy Efficient Ethernet
- IEEE 802.1Q for VLAN frame format

Feature List

- Data rates of 10/100/1000 Mbit/s through an external PHY interface
- Communication with an external Ethernet PHY through IEEE 802.3-compliant MII, RMII, or RGMII interface (only one can be used at a time)
- Full-duplex and half-duplex modes
 - Carrier Sense Multiple Access or Collision Detection (CSMA/CD) protocol in half-duplex mode
 - IEEE 802.3x flow control in full-duplex mode
 - Optional forwarding of received pause control frame to the user application in full-duplex mode
 - Back-pressure flow control in half-duplex mode
 - Automatic transmission of zero-quanta pause frame on deassertion of flow control input in full-duplex mode
- Preamble and start-of-frame data (SFD) insertion in Transmit, and deletion in Receive paths
- Automatic CRC and padding (all 0) generation controllable on a per-frame basis
- Options for automatic padding generation for data below the minimum frame length

- Programmable frame length supporting jumbo frames of up to 16 KB
- Programmable inter-frame gap (IFG) from 40 to 96 bit times in steps of 8
- Flexible address filtering modes:
 - Up to nine 48-bit perfect address filters with per-byte masking
 - Up to nine 48-bit source address (SA) comparisons with per-byte masking
 - Option to pass all multicast addressed frames
 - Promiscuous mode to pass all frames without filtering for network monitoring
 - Passes all incoming packets (as per filter) with a status report
- Separate 32-bit status returned for transmission and reception packets
- IEEE 802.1Q VLAN tag detection for reception frames
- Separate transmission, reception, and control interfaces for the application
- Management Data Input/Output (MDIO) interface for PHY device configuration and management
- Checksum offload for received IPv4 and TCP packets encapsulated by the Ethernet frame
- Checking IPv4 header checksum and TCP, UDP, or ICMP checksum encapsulated in IPv4 or IPv6 datagrams
- 64-bit timestamp for each transmitted and received frame (see IEEE 1588-2008)
- Energy Efficient Ethernet support (see IEEE 802.3az-2010)
- CRC replacement, SA insertion/replacement, and VLAN insertion/replacement/deletion in transmit frames
- Two FIFOs: 1024-byte TX FIFO and 256-byte RX FIFO
- Receive status vectors inserted into RX FIFO after the EOF (end of frame) transfer, allowing multiple-frame storage without requiring an additional FIFO for status
- Option to forward good runt frames
- Statistics generation with pulse signaling for dropped or corrupted frames due to RX FIFO overflow
- Automatic re-transmission of collision frames
- Frame discarding in cases of late collisions, excessive collisions, excessive deferrals, or underflow conditions
- Software control for TX FIFO flushing

Pin Assignment

EMAC RGMII interface corresponds to the following designated pins in IO MUX:

- The pins are multiplexed with GPIO8–GPIO19

EMAC RMII interface corresponds to the following designated pins in IO MUX:

- The pins are multiplexed with GPIO8, GPIO9, GPIO12, GPIO13, GPIO15, GPIO18, GPIO19

EMAC MII interface uses the available RGMII pins and additionally requires three arbitrary GPIOs for signals such as rxderr/csr/col.

MDIO and other signals can be routed to any GPIOs via the GPIO Matrix.

4.2.2.9 CAN FD Controller

The CAN FD is a multi-master multi-cast communication protocol. The CAN FD controller facilitates the communication based on this protocol. The CAN FD controllers integrated in ESP32-S31 are protocol-compatible with the CAN FD specification but have not been formally certified.

Feature List

- compliant with ISO11898-1:2015
- RX buffer FIFO with 128 words (6 CAN FD frames with 64-byte data payloads, 21 CAN/CAN FD frames with 8-byte data payloads)
- 4 TX buffers (1 CAN FD frame in each TX buffer)
- 32-bit APB interface
- support of ISO and non-ISO CAN FD protocol
- timestamping and time triggered transmission
- three single-bit filters and one range filter
- support interrupts
- loopback, bus monitoring, ACK forbidden, self test, and restricted operation modes

Pin Assignment

The pins for the CAN FD Controller can be chosen from any GPIOs via the GPIO Matrix.

4.2.2.10 SD/MMC Host Controller (SDHOST)

ESP32-S31 has an SD/MMC Host Controller.

Feature List

- Supports two external cards
- SD Memory Card Specifications v3.0 and v3.01
- Secure Digital I/O (SDIO) v3.0
- CE-ATA v1.1
- MMC v4.41 and eMMC v4.5, v4.51
- 1-bit and 4-bit bus width modes (8-bit mode not supported)

ESP32-S31 SD/SDIO/MMC Host Controller can simultaneously support two SD/SDIO/MMC 4.41 cards, or support only one SD card operating at 1.8 V.

Pin Assignment

For the SD/SDIO/MMC host controller, card one can use GPIO20–GPIO25 via IO MUX, and card two can use GPIO35–GPIO40 via IO MUX.

4.2.2.11 LED PWM Controller (LEDC)

The LED PWM Controller is a peripheral designed to generate PWM signals for LED control. It has specialized features such as automatic duty cycle fading. However, the LED PWM Controller can also be used to generate PWM signals for other purposes.

Feature List

- Two independent LED PWM controllers, each with eight independent PWM generator channels (16 channels in total)
- Maximum PWM duty cycle resolution: 20 bits
- Four independent timers per controller with 20-bit counters, fractional clock divider, and configurable overflow value
- Adjustable phase of PWM signal output
- PWM duty cycle dithering
- Automatic duty cycle fading —gradual increase/decrease of a PWM's duty cycle without interference from the processor. An interrupt will be generated upon fade completion
- Up to 16 duty cycle ranges for each PWM generator to generate gamma curve signals; each range can be independently configured for fade direction (increase or decrease), fade amount (per-step duty change), fade count (number of steps per range), and fade frequency
- PWM signal output in low-power mode (Light-sleep mode)
- Event generation and task response related to the Event Task Matrix (ETM) peripheral

Pin Assignment

The pins for the LED PWM controller can be chosen from any GPIOs via the GPIO Matrix.

4.2.2.12 Motor Control PWM (MCPWM)

ESP32-S31 integrates four Motor Control PWM (MCPWM) controllers (MCPWM0, MCPWM1, MCPWM2, and MCPWM3) that can be used to drive digital motors and smart light.

Feature List

- Each MCPWM has a clock divider (prescaler), three PWM timers, three PWM operators, an Event Task Matrix (ETM) module, a Fault Detection module, and a dedicated capture module. PWM timers are used to generate timing references. PWM operators generate desired waveform based on the timing references
- A PWM operator can use the timing reference of any PWM timer

- A PWM operator can use the same timing reference with other PWM operators
- PWM operators can use different PWM timers' values to produce independent PWM signals
- PWM timers can be synchronized

Pin Assignment

The pins for the Motor Control PWM can be chosen from any GPIOs via the GPIO Matrix.

4.2.2.13 Remote Control Peripheral (RMT)

The Remote Control Peripheral (RMT) supports four channels of infrared remote transmission and four channels of infrared remote reception. By controlling pulse waveform through software, it supports various infrared and other single wire protocols.

Feature List

- Eight channels:
 - TX channels 0–3
 - RX channels 4–7
 - Eight channels share a 384 x 32-bit RAM
- The transmitter supports:
 - Normal TX mode
 - Wrap TX mode
 - Continuous TX mode
 - Modulation on TX pulses
 - Multiple channels transmitting data simultaneously (programmable)
 - GDMA access supported by TX channel 3
- The receiver supports:
 - Normal RX mode
 - Wrap RX mode
 - RX filtering
 - Demodulation on RX pulses
 - GDMA access supported by RX channel 7

Pin Assignment

The pins for the remote control peripheral can be chosen from any GPIOs via the GPIO Matrix.

4.2.2.14 Parallel IO Controller (PARLIO)

ESP32-S31 contains a Parallel IO controller (PARLIO) capable of transferring data between external devices and internal memory on a parallel bus through General Direct Memory Access (GDMA).

Feature List

- Various clock sources:
 - Including external IO clock PAD_CLK_TX/RX and internal system clock XTAL_CLK, PLL_F160M_CLK, and RC_FAST_CLK
 - Maximum IO clock frequency of 40 MHz
 - Integer and fractional clock frequency division
- 1/2/4/8/16-bit configurable data bus width
- Full-duplex communication with 16-bit data bus width
- Bit reversal when data bus width is 1/2/4-bit
- RX unit for receiving IO parallel data, which supports:
 - Output clock gating
 - RX unit input and output clock inverse
 - Various receive modes
 - Configurable GDMA SUC EOF generation
 - Configurable IO pin of external enable signal
- TX unit for sending IO parallel data, which supports:
 - Output clock gating
 - TX unit input and output clock inverse
 - Configurable TX EOF generation
 - Valid signal output
 - Configurable bus idle value

Pin Assignment

The pins for the parallel IO controller can be chosen from any GPIOs via the GPIO Matrix.

4.2.2.15 BitScrambler

The ESP32-S31 has an extensive amount of DMA-capable peripherals. These can move data from memory to an external device, and vice versa, without any interference from the CPU. This only works if the external device needs or emits the data in question in the same format as the software expects it: if not, the CPU needs to rewrite the format of the data. Examples include a need to swap bytes, reverse bytes, and shift the data left or right.

Since bitwise operations can be relatively CPU-intensive and DMA is designed specifically to offload such work from the CPU, ESP32-S31 integrates two dedicated peripherals called BitScramblers. These modules are designed to transform data formats during transfers between memory and peripherals. One BitScrambler handles memory-to-peripheral (or memory-to-memory) transfers, while the other is dedicated to peripheral-to-memory transfers. While BitScramblers can handle the bitwise operations mentioned earlier, they are in fact flexible, programmable state machines capable of performing more advanced transformations as well.

Feature List

- Two BitScramblers, one for RX (peripheral-to-memory), one for TX (memory-to-peripheral)
- Support for memory-to-memory transfers
- Processing up to 32 bits per DMA clock period
- Data path controlled by a BitScrambler program stored in instruction memory
- Input registers able to read 0, 8, 16, or 32 bits per clock cycle
- Output registers:
 - Able to write 0, 8, 16, or 32 bits per clock cycle
 - Data sources for output register bits: 64 bits of input data, two counters, LUT RAM data, data output of last cycle, comparators
 - With some restrictions, each of the 32 output register bits can come from any bit on the data sources
- An 8 x 257-bit instruction memory for storing eight instructions, controlling control flow, and the data path
- 2048 bytes of lookup table (LUT) memory, configurable as various word widths

Pin Assignment

The BitScrambler does not directly interact with IOs, so it has no pins assigned.

4.2.3 Analog Signal Processing

This subsection describes components on the chip that sense and process real-world data.

4.2.3.1 Touch Sensor (TOUCH)

ESP32-S31 has 14 capacitive touch sensor units, each of which can be connected to an external touch panel via a touch pin (GPIO pin) to constitute a touch sensor system. The ESP32-S31 touch sensor system is mainly used in human-computer interactions to detect finger touch or proximity. It also supports frequency hopping sampling and features moisture resistance and waterproof design.

Feature List

- Detection of 14 capacitive touch pins
- Sampling triggered by software or dedicated hardware timer
- Two sampling methods:

- Pulses from the touch pins used as clock signals to count the sampling period
- Pulses from the touch pins used as digital signals; sample the rising edge of the digital signal with the system clock to count the sampling period
- Scan mode, supporting sequential sampling of multiple touch pins by configuring the Touch FSM.
- Timeout mechanism to monitor channel abnormality
- Frequency hopping to increase the anti-interference of detection
- Proximity sensing mode with up to three configurable channels
- Configuration of individual touch sensors to operate normally in sleep mode
- Wake-up by touch sensor
- Moisture resistance
- Waterproof design

Pin Assignment

The touch sensor interface is multiplexed with GPIO6~GPIO19 pins. When the pins are configured for the analog function, the multiplexed digital functions are disabled.

4.2.3.2 Temperature Sensor (TSENS)

ESP32-S31 provides a temperature sensor for real-time monitoring of temperature changes within the chip. The sensor converts analog voltage to digital values and provides compensation for temperature offsets.

Feature List

- Software-triggered temperature measurement, which once triggered, the sensor continuously measures temperature. Software can read the data at any time.
- Hardware-triggered automatic temperature monitoring
- Two automatic monitoring modes with interrupt generation
- Configurable temperature offset based on the application scenario for improved accuracy
- Configurable temperature measurement range
- Support for Event Task Matrix (ETM)-related events and tasks

4.2.3.3 ADC Controller (ADC)

ESP32-S31 integrates two 12-bit successive approximation ADCs (SAR ADCs), each capable of measuring analog signals from up to eight pins.

Feature List

- 12-bit resolution
- Analog inputs sampling from up to eight pins per ADC (16 pins in total)

- One-shot sampling mode and multi-channel sampling mode
- In multi-channel sampling mode:
 - Configurable channel sampling sequence
 - Two filters with configurable filter coefficients
 - Two threshold monitors that can trigger an interrupt when the filtered value is below a low threshold or above a high threshold
 - Continuous transfer of converted data to memory via GDMA interface
- Support for several Event Task Matrix (ETM) related events and tasks

Pin Assignment

SAR ADC1 pins are multiplexed with GPIO42 ~ GPIO49; SAR ADC2 pins are multiplexed with GPIO50 ~ GPIO57.

4.2.3.4 DAC Controller (DAC)

ESP32-S31 provides a digital-to-analog converter that converts digital signals into analog voltages and outputs them on two dedicated chip pads. The two channels connect to two pads respectively and can output independently without affecting each other.

Feature List

- Supports output value configuration by software via GDMA, or sine-wave output via an internal lookup table
- Output voltage range: 0 V to 3.3 V
- Supports maintaining output level in ultra-low-power mode

Pin Assignment

DAC uses dedicated pads and is supported only on GPIO4 (DAC channel 0) and GPIO5 (DAC channel 1).

4.2.3.5 Analog Voltage Comparator

ESP32-S31 includes one analog voltage comparator with four dedicated pads. Three of these pads can be selected to compare their voltages against the remaining pad. Alternatively, any of the three pads can be compared against an internally adjustable stable reference voltage.

Feature List

- Reference voltage selectable between internal reference and external reference
- Internal reference voltage range: 0 ~ 0.7 * VDDPST
- Internal reference voltage supports hysteresis
- ETM support
- Interrupt output when the measured voltage crosses the reference voltage

Pin Assignment

The analog voltage comparator uses dedicated pads and is supported only on GPIO37, GPIO38, GPIO39, and GPIO40. Any of these pads can be configured as the measured-input pad or the reference-input pad.

4.3 Wireless Communication

This section describes the chip's wireless communication capabilities, spanning radio technology, Wi-Fi, Bluetooth LE, Bluetooth Classic and 802.15.4.

4.3.1 Radio

This subsection describes the fundamental radio technology embedded in the chip that facilitates wireless communication and data exchange.

4.3.1.1 2.4 GHz Receiver

The 2.4 GHz receiver demodulates the 2.4 GHz RF signal to quadrature baseband signals and converts them to the digital domain with two high-resolution, high-speed ADCs. To adapt to varying signal channel conditions, ESP32-S31 integrates RF filters, Automatic Gain Control (AGC), DC offset cancelation circuits, and baseband filters.

4.3.1.2 2.4 GHz Transmitter

The 2.4 GHz transmitter modulates the quadrature baseband signals to the 2.4 GHz RF signal, and drives the antenna with a high-powered CMOS power amplifier. The use of digital calibration further improves the linearity of the power amplifier.

Additional calibrations are integrated to cancel any radio imperfections, such as:

- Carrier leakage
- I/Q amplitude/phase matching
- Baseband nonlinearities
- RF nonlinearities
- Antenna matching

These built-in calibration routines reduce the cost, time, and specialized equipment required for product testing.

4.3.1.3 Clock Generator

The clock generator produces quadrature clock signals of 2.4 GHz for both the receiver and the transmitter. All components of the clock generator are integrated into the chip, including inductors, varactors, filters, regulators and dividers.

The clock generator has built-in calibration and self-test circuits. Quadrature clock phases and phase noise are optimized on chip with patented calibration algorithms which ensure the best performance of the receiver and the transmitter.

4.3.2 Wi-Fi

This subsection describes the chip's Wi-Fi capabilities, which facilitate wireless communication at a high data rate.

4.3.2.1 Wi-Fi Radio and Baseband

The ESP32-S31 Wi-Fi radio and baseband support the following features:

- 1T1R in 2.4 GHz band
- 802.11ax
 - 20 MHz-only non-AP mode
 - MCS0 ~MCS9
 - Uplink and downlink OFDMA
 - Downlink MU-MIMO (multi-user, multiple input, multiple output)
 - Longer OFDM symbol, with 0.8, 1.6, 3.2 μ s guard interval
 - DCM (dual carrier modulation), up to 16-QAM
 - Single-user/multi-user beamformee
 - Channel quality indication (CQI)
 - RX STBC (single spatial stream)
- 802.11b/g/n
 - MCS0 ~MCS7 that supports 20 MHz and 40 MHz bandwidth
 - MCS32
 - Data rate up to 150 Mbps
 - 0.4 μ s guard interval
- Adjustable transmitting power
- Antenna diversity

ESP32-S31 supports antenna diversity with an external RF switch. This switch is controlled by one or more GPIOs, and used to select the best antenna to minimize the effects of channel imperfections.

4.3.2.2 Wi-Fi MAC

ESP32-S31 implements the full IEEE 802.11 b/g/n/ax Wi-Fi MAC protocol. ESP32-S31 supports the Basic Service Set (BSS) STA and SoftAP operations under the Distributed Coordination Function (DCF). Power management is handled automatically with minimal host interaction to minimize the active duty period.

The ESP32-S31 Wi-Fi MAC applies the following low-level protocol functions automatically:

- Four virtual Wi-Fi interfaces
- Infrastructure BSS in Station mode, SoftAP mode, Station + SoftAP mode, and promiscuous mode
- RTS protection, CTS-to-Self protection, Immediate Block ACK
- Fragmentation and defragmentation
- TX/RX A-MPDU, TX/RX A-MSDU
- Transmit opportunity (TXOP)

- Wi-Fi multimedia (WMM)
- GCMP, CCMP, TKIP, WAPI, WEP, BIP, WPA2-PSK/WPA2-Enterprise, and WPA3-PSK/WPA3-Enterprise
- Automatic beacon monitoring (hardware TSF)
- 802.11mc FTM
- 802.11ax supports:
 - Target wake time (TWT) requester
 - Multiple BSSIDs
 - Triggered response scheduling
 - Multi-user Request-to-Send (MU-RTS), Multi-user Block ACK Request (MU-BAR), and Multi-STA Block ACK (M-BA) frame
 - Intra-PPDU power saving mechanism
 - Two network allocation vectors (NAV)
 - BSS coloring
 - Spatial reuse
 - Uplink power headroom
 - Operating mode control
 - Buffer status report
 - TXOP duration RTS threshold
 - UL-OFDMA random access (UORA)

4.3.2.3 Networking Features

Espressif provides libraries for TCP/IP networking, ESP-WIFI-MESH networking, and other networking protocols over Wi-Fi. TLS 1.0, 1.1 and 1.2 is also supported.

4.3.3 Bluetooth LE

This subsection describes the chip's Bluetooth LE capabilities for low-power data communication, supporting a wide range of use cases such as efficient data transfer, Bluetooth LE Audio, positioning, and electronic shelf labels.

4.3.3.1 Bluetooth LE PHY

Bluetooth Low Energy PHY in ESP32-S31 supports:

- 1 Mbps PHY
- 2 Mbps PHY for Higher Data Rates
- Coded PHY for Longer Range (125 Kbps and 500 Kbps)
- HW Listen Before Talk (LBT)

4.3.3.2 Bluetooth LE Link Controller

Bluetooth Low Energy Link Controller and Host in ESP32-S31 support:

- LE Audio (Isochronous Channels, BIS and CIS)
- Direction Finding (AoA/AoD)
- Periodic Advertising with Responses (PAWR)
- LE Connection Subrating
- LE Advertising Extensions and Multiple Advertising Sets
- Simultaneous Operation of Broadcaster, Observer, Central, and Peripheral Devices
- Adaptive Frequency Hopping and Channel Assessment
- LE Channel Selection Algorithm #2
- LE Power Control
- Advertising Coding Selection
- Encrypted Advertising Data
- LE GATT Security Levels Characteristic
- AdvDataInfo in Periodic Advertising
- LE Channel Classification
- Enhanced Attribute Protocol
- Advertising Channel Index
- GATT Caching
- Periodic Advertising Sync Transfer
- High Duty Cycle Non-Connectable Advertising
- LE Data Packet Length Extension
- LE Secure Connections
- LE Privacy 1.2
- Link Layer Extended Scanner Filter Policies
- Low Duty Cycle Directed Advertising
- Link Layer Encryption
- LE Ping

4.3.4 Bluetooth Classic

This subsection describes the chip's Bluetooth Classic capabilities for sustained data throughput and traditional Bluetooth audio (e.g., A2DP, HFP), suitable for established audio ecosystems and continuous data links.

4.3.4.1 Bluetooth Classic PHY

ESP32-S31 series chips Bluetooth Classic physical layer supports the following features:

- Basic rate (GFSK): 1 Mbps
- Enhanced data rate ($\pi/4$ DQPSK): 2 Mbps
- Enhanced data rate (8DPSK): 3 Mbps
- Hardware-implemented clear channel assessment (CCA)
- Maximum output power meets Power Class 1 device requirements

4.3.4.2 Bluetooth Classic Link Controller

ESP32-S31 series chips Bluetooth Classic link controller can perform device inquiry and paging operations, can form piconets and scatternets, and supports active mode and sniff mode in connection state, as well as role switching, data encryption, power control, adaptive frequency hopping and other operations. The main features of the Bluetooth Classic link controller are as follows:

- Device discovery (inquiry and inquiry scan)
- Connection establishment (paging and paging scan)
- Asynchronous connection-oriented (ACL) and synchronous connection-oriented (SCO/eSCO)
- Voice coding supports A-law, μ -law, CVSD and transparent data
- Voice data channel supports HCI or PCM/I2S interface
- Secure simple pairing (SSP)
- EO encryption and AES-CCM encryption
- Secure Connections
- Channel classification and adaptive frequency hopping (AFH)
- Sniff mode and Sniff Subrating
- Role switching
- Traditional power control and enhanced power control
- Ping
- Piconet and scatternet management
- Active Peripheral Broadcast

4.3.5 802.15.4

This subsection describes the chip's compatibility with the 802.15.4 standard, which facilitates wireless communication for low-power, short-range applications.

4.3.5.1 802.15.4 PHY

ESP32-S31's 802.15.4 PHY supports:

- O-QPSK PHY in 2.4 GHz
- 250 Kbps data rate
- RSSI and LQI

4.3.5.2 802.15.4 MAC

ESP32-S31 supports most key features defined in [IEEE Standard 802.15.4-2015](#), including:

- CSMA/CA
- Active scan and energy detect
- HW frame filter
- HW auto acknowledge
- HW auto frame pending
- Coordinated sampled listening (CSL)

5 Electrical Characteristics

Note:

The values presented in this section are **preliminary** and may change with the final release of this datasheet.

5.1 Absolute Maximum Ratings

Stresses above those listed in Table 5-1 *Absolute Maximum Ratings* may cause permanent damage to the device. These are stress ratings only and normal operation of the device at these or any other conditions beyond those indicated in Section 5.2 *Recommended Power Supply Characteristics* is not implied. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability.

Table 5-1. Absolute Maximum Ratings

Parameter	Description	Min	Max	Unit
Input power pins ¹	Allowed input voltage	−0.3	3.6	V
I_{output} ²	Cumulative IO output current	—	1500	mA
T_{STORE}	Storage temperature	−40	150	°C

¹ For more information on input power pins, see Section 2.5.1 *Power Pins*.

² The product proved to be fully functional after all its IO pins were pulled high while being connected to ground for 24 consecutive hours at ambient temperature of 25 °C.

5.2 Recommended Power Supply Characteristics

For recommended ambient temperature, see Section 1 *ESP32-S31 Series Information*.

Table 5-2. Recommended Power Characteristics

Parameter ¹	Description	Min	Typ	Max	Unit
VDDA1, VDDA2, VDDA3, VDDA4	Recommended input voltage	3.0	3.3	3.6	V
VDDPST_1, VDDPST_2, VDDPST_3, VDDPST_4 ²	Recommended input voltage	3.0	3.3	3.6	V
I_{VDD}	Cumulative input current	0.6	—	—	A

¹ See in conjunction with Section 2.5 *Power Supply*.

² When writing to eFuses, the voltage should not exceed 3.3 V because eFuse programming circuitry is sensitive to higher voltages.

5.3 VDD_SPI Output Characteristics

Table 5-3. VDD_SPI Internal and Output Characteristics

Parameter	Description ¹	Typ	Unit
R_{SPI}	VDD_SPI powered by VDD3P3 via R_{SPI} for 3.3 V flash/PSRAM ²	3	Ω
I_{SPI}	Output current when VDD_SPI is powered by Voltage Regulator for 1.8 V flash/PSRAM	100	mA

¹ See in conjunction with Section 2.5.2 *Power Scheme*.

² VDD3P3 must be more than $VDD_flash_min + I_flash_max \times R_{SPI}$;
where

- VDD_flash_min – minimum operating voltage of flash/PSRAM
- I_flash_max – maximum operating current of flash/PSRAM

5.4 DC Characteristics (3.3 V, 25 °C)

Table 5-4. DC Characteristics (3.3 V, 25 °C)

Parameter	Description	Min	Typ	Max	Unit
C_{IN}	Pin capacitance	—	2	—	pF
V_{IH}	High-level input voltage	$0.75 \times VDD^1$	—	$VDD^1 + 0.3$	V
V_{IL}	Low-level input voltage	−0.3	—	$0.25 \times VDD^1$	V
I_{IH}	High-level input current	—	—	50	nA
I_{IL}	Low-level input current	—	—	50	nA
V_{OH}^2	High-level output voltage	$0.8 \times VDD^1$	—	—	V
V_{OL}^2	Low-level output voltage	—	—	$0.1 \times VDD^1$	V
I_{OH}	High-level source current ($VDD^1 = 3.3$ V, $V_{OH} \geq 2.64$ V, PAD_DRIVER = 3)	—	40	—	mA
I_{OL}	Low-level sink current ($VDD^1 = 3.3$ V, $V_{OL} = 0.495$ V, PAD_DRIVER = 3)	—	28	—	mA
R_{PU}	Internal weak pull-up resistor	—	45	—	k Ω
R_{PD}	Internal weak pull-down resistor	—	45	—	k Ω
V_{IH_nRST}	Chip reset release voltage (CHIP_PU voltage is within the specified range)	$0.75 \times VDD^1$	—	$VDD^1 + 0.3$	V
V_{IL_nRST}	Chip reset voltage (CHIP_PU voltage is within the specified range)	−0.3	—	$0.15 \times VDD^1$	V

¹ VDD – voltage from a power pin of a respective power domain.

² V_{OH} and V_{OL} are measured using high-impedance load.

5.5 Current Consumption

5.5.1 RF Current Consumption in Active Mode

The current consumption measurements are taken with a 3.3 V supply at 25 °C ambient temperature.

TX current consumption is rated at a 100% duty cycle.

RX current consumption is rated when the peripherals are disabled and the CPU idle.

Table 5-5. Current Consumption for Wi-Fi (2.4 GHz) in Active Mode

Work Mode	RF Condition	Description	Peak (mA)
Active (RF working)	TX	802.11b, 1 Mbps, DSSS @ 21 dBm	377
		802.11g, 54 Mbps, OFDM @ 18 dBm	297
		802.11n, HT20, MCS7 @ 18 dBm	299
		802.11n, HT40, MCS7 @ 17 dBm	300
		802.11ax, MCS9 @ 15 dBm	265
	RX	802.11b/g/n, HT20	110
		802.11n, HT40	117
		802.11ax, HE20	112

Table 5-6. Current Consumption for Bluetooth LE in Active Mode

Work Mode	RF Condition	Description	Peak (mA)
Active (RF working)	TX	Bluetooth LE @ 20 dBm	340
		Bluetooth LE @ 11 dBm	225
		Bluetooth LE @ 0 dBm	160
		Bluetooth LE @ -17.5 dBm	120
	RX	Bluetooth LE	105

Table 5-7. Current Consumption for Bluetooth Classic in Active Mode

Work Mode	RF Condition	Description	Peak (mA)
Active (RF working)	TX	Bluetooth Classic @ 12 dBm	225
		Bluetooth Classic @ 6 dBm	185
		Bluetooth Classic @ 0 dBm	145
		Bluetooth Classic @ -17.5 dBm	116
	RX	Bluetooth Classic	96

Table 5-8. Current Consumption for 802.15.4 in Active Mode

Work Mode	RF Condition	Description	Peak (mA)
Active (RF working)	TX	802.15.4 @ 19 dBm	310
		802.15.4 @ 13 dBm	230
		802.15.4 @ 1 dBm	170

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Table 5-8 – cont'd from previous page

Work Mode	RF Condition	Description	Peak (mA)
		802.15.4 @ -17.5 dBm	120
	RX	802.15.4	104

5.5.2 Current Consumption in Other Modes

Table 5-9. Current Consumption in Modem-sleep Mode

Mode	CPU Frequency (MHz)	Description	Typ (mA)	
			All Peripherals Clocks Disabled	All Peripherals Clocks Enabled ¹
Modem-sleep ^{2,3}	320	Dual-core WAITI	25	75
		Dual-core CoreMark	91	147
		Core 0 WAITI, core 1 SIMD	64	114
		Core 0 CoreMark, core 1 SIMD	100	152
	240	Dual-core WAITI	23	72
		Dual-core CoreMark	74	126
		Core 0 WAITI, core 1 SIMD	52	102
		Core 0 CoreMark, core 1 SIMD	80	130
	160	Dual-core WAITI	21	70
		Dual-core CoreMark	57	107
		Core 0 WAITI, core 1 SIMD	41	90
		Core 0 CoreMark, core 1 SIMD	59	109
	80	Dual-core WAITI	19	68
		Dual-core CoreMark	37	86
		Core 0 WAITI, core 1 SIMD	29	78
		Core 0 CoreMark, core 1 SIMD	38	87
	40	Dual-core WAITI	13	56
		Dual-core CoreMark	22	65
		Core 0 WAITI, core 1 SIMD	18	61
		Core 0 CoreMark, core 1 SIMD	23	67

¹ In practice, the current consumption might be different depending on which peripherals are enabled.

² In Modem-sleep mode, Wi-Fi is clock gated.

³ In Modem-sleep mode, the consumption might be higher when accessing flash.

Table 5-10. Current Consumption in Low-Power Modes

Mode	Description	Typ (mA)
Light-sleep ¹	All GPIOs are high-impedance, and all power supplies are enabled	1
	All GPIOs are high-impedance, most of peripherals are disabled, and chip is connected through USB	0.07
	All peripherals are disabled, and data is stored in HP memory	0.05
Deep-sleep	LP timer and LP memory are powered on	0.012
Power off	CHIP_PU is set to low level, the chip is powered off	0.001

¹ The current in Light-sleep mode refers to the current measured when the PSRAM is not powered. In Light-sleep mode, if the PSRAM is powered on, the chip's internal current increases by about 0.17 mA, in addition to the current required for the PSRAM's operating mode.

5.6 Memory Specifications

The data below is sourced from the memory vendor datasheet. These values are guaranteed through design and/or characterization but are not fully tested in production. Devices are shipped with the memory erased.

Table 5-11. Flash Specifications

Parameter	Description	Min	Typ	Max	Unit
VCC	Power supply voltage (1.8 V)	1.65	1.80	2.00	V
	Power supply voltage (3.3 V)	2.7	3.3	3.6	V
F _C	Maximum clock frequency	80	—	—	MHz
—	Program/erase cycles	100,000	—	—	cycles
T _{RET}	Data retention time	20	—	—	years
T _{PP}	Page program time	—	0.8	5	ms
T _{SE}	Sector erase time (4 KB)	—	70	500	ms
T _{BE1}	Block erase time (32 KB)	—	0.2	2	s
T _{BE2}	Block erase time (64 KB)	—	0.3	3	s
T _{CE}	Chip erase time (16 Mb)	—	7	20	s
	Chip erase time (32 Mb)	—	20	60	s
	Chip erase time (64 Mb)	—	25	100	s
	Chip erase time (128 Mb)	—	60	200	s
	Chip erase time (256 Mb)	—	70	300	s

Table 5-12. PSRAM Specifications

Parameter	Description	Min	Typ	Max	Unit
VCC	Power supply voltage (1.8 V)	1.62	1.80	1.98	V
	Power supply voltage (3.3 V)	2.7	3.3	3.6	V
F _C	Maximum clock frequency	80	—	—	MHz

6 RF Characteristics

This section contains tables with RF characteristics of the Espressif product.

The RF data is measured at the antenna port, where RF cable is connected, including the front-end loss. The front-end circuit is a 0 Ω resistor.

Devices should operate in the center frequency range allocated by regional regulatory authorities. The target center frequency range and the target transmit power are configurable by software. See [ESP RF Test Tool and Test Guide](#) for instructions.

Unless otherwise stated, the RF tests are conducted with a 3.3 V ($\pm 5\%$) supply at 25 °C ambient temperature.

6.1 Wi-Fi Radio

Table 6-1. 2.4 GHz Wi-Fi RF Characteristics

Name	Description
Center frequency range of operating channel	2412 ~ 2484 MHz
Wi-Fi wireless standard	IEEE 802.11b/g/n/ax

6.1.1 Wi-Fi RF Transmitter (TX) Characteristics

Table 6-2. 2.4 GHz TX Power with Spectral Mask and EVM Meeting 802.11 Standards

Rate	Min (dBm)	Typ (dBm)	Max (dBm)
802.11b, 1 Mbps, DSSS	—	21.0	—
802.11b, 11 Mbps, CCK	—	21.0	—
802.11g, 6 Mbps, OFDM	—	20.0	—
802.11g, 54 Mbps, OFDM	—	18.0	—
802.11n, HT20, MCS0	—	20.0	—
802.11n, HT20, MCS7	—	18.0	—
802.11n, HT40, MCS0	—	19.0	—
802.11n, HT40, MCS7	—	17.0	—
802.11ax, HE20, MCS0	—	20.0	—
802.11ax, HE20, MCS9	—	15.0	—

Table 6-3. 2.4 GHz TX EVM Test¹

Rate	Min (dB)	Typ (dB)	Limit (dB)
802.11b, 1 Mbps, DSSS	—	-25.0	-10.0
802.11b, 11 Mbps, CCK	—	-25.0	-10.0

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Table 6-3 – cont'd from previous page

Rate	Min (dB)	Typ (dB)	Limit (dB)
802.11g, 6 Mbps, OFDM	—	–24.0	–5.0
802.11g, 54 Mbps, OFDM	—	–30.0	–25.0
802.11n, HT20, MCS0	—	–22.0	–5.0
802.11n, HT20, MCS7	—	–30.0	–27.0
802.11n, HT40, MCS0	—	–26.0	–5.0
802.11n, HT40, MCS7	—	–31.0	–27.0
802.11ax, HE20, MCS0	—	–23.0	–5.0
802.11ax, HE20, MCS9	—	–34.0	–32.0

¹ EVM is measured at the corresponding typical TX power provided in Table 6-2 *2.4 GHz TX Power with Spectral Mask and EVM Meeting 802.11 Standards* above.

6.1.2 Wi-Fi RF Receiver (RX) Characteristics

For RX tests, the PER (packet error rate) limit is 8% for 802.11b, and 10% for 802.11g/n/ax.

Table 6-4. 2.4 GHz RX Sensitivity

Rate	Min (dBm)	Typ (dBm)	Max (dBm)
802.11b, 1 Mbps, DSSS	—	–99.5	—
802.11b, 2 Mbps, DSSS	—	–96.5	—
802.11b, 5.5 Mbps, CCK	—	–93.5	—
802.11b, 11 Mbps, CCK	—	–89.5	—
802.11g, 6 Mbps, OFDM	—	–94.5	—
802.11g, 9 Mbps, OFDM	—	–92.5	—
802.11g, 12 Mbps, OFDM	—	–92.5	—
802.11g, 18 Mbps, OFDM	—	–89.5	—
802.11g, 24 Mbps, OFDM	—	–87.0	—
802.11g, 36 Mbps, OFDM	—	–83.0	—
802.11g, 48 Mbps, OFDM	—	–79.0	—
802.11g, 54 Mbps, OFDM	—	–77.5	—
802.11n, HT20, MCS0	—	–94.5	—
802.11n, HT20, MCS1	—	–92.0	—
802.11n, HT20, MCS2	—	–89.5	—
802.11n, HT20, MCS3	—	–86.0	—
802.11n, HT20, MCS4	—	–83.0	—
802.11n, HT20, MCS5	—	–78.5	—
802.11n, HT20, MCS6	—	–77.0	—
802.11n, HT20, MCS7	—	–75.0	—
802.11n, HT40, MCS0	—	–92.0	—
802.11n, HT40, MCS1	—	–89.5	—

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Table 6-4 – cont'd from previous page

Rate	Min (dBm)	Typ (dBm)	Max (dBm)
802.11n, HT40, MCS2	—	–87.0	—
802.11n, HT40, MCS3	—	–83.5	—
802.11n, HT40, MCS4	—	–80.5	—
802.11n, HT40, MCS5	—	–76.0	—
802.11n, HT40, MCS6	—	–74.5	—
802.11n, HT40, MCS7	—	–73.5	—
802.11ax, HE20, MCS0	—	–94.0	—
802.11ax, HE20, MCS1	—	–91.0	—
802.11ax, HE20, MCS2	—	–88.0	—
802.11ax, HE20, MCS3	—	–85.5	—
802.11ax, HE20, MCS4	—	–82.0	—
802.11ax, HE20, MCS5	—	–78.0	—
802.11ax, HE20, MCS6	—	–76.5	—
802.11ax, HE20, MCS7	—	–74.5	—
802.11ax, HE20, MCS8	—	–71.0	—
802.11ax, HE20, MCS9	—	–68.5	—

Table 6-5. 2.4 GHz Maximum RX Level

Rate	Min (dBm)	Typ (dBm)	Max (dBm)
802.11b, 1 Mbps, DSSS	—	5	—
802.11b, 11 Mbps, CCK	—	5	—
802.11g, 6 Mbps, OFDM	—	5	—
802.11g, 54 Mbps, OFDM	—	0	—
802.11n, HT20, MCS0	—	5	—
802.11n, HT20, MCS7	—	0	—
802.11n, HT40, MCS0	—	5	—
802.11n, HT40, MCS7	—	0	—
802.11ax, HE20, MCS0	—	5	—
802.11ax, HE20, MCS9	—	0	—

Table 6-6. 2.4 GHz RX Adjacent Channel Rejection

Rate	Min (dB)	Typ (dB)	Max (dB)
802.11b, 1 Mbps, DSSS	—	43	—
802.11b, 11 Mbps, CCK	—	40	—
802.11g, 6 Mbps, OFDM	—	33	—
802.11g, 54 Mbps, OFDM	—	18	—

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Table 6-6 – cont'd from previous page

Rate	Min (dB)	Typ (dB)	Max (dB)
802.11n, HT20, MCS0	—	30	—
802.11n, HT20, MCS7	—	15	—
802.11n, HT40, MCS0	—	23	—
802.11n, HT40, MCS7	—	10	—
802.11ax, HE20, MCS0	—	25	—
802.11ax, HE20, MCS9	—	2	—

6.2 Bluetooth LE Radio

Table 6-7. Bluetooth LE RF Characteristics

Name	Description
Center frequency range of operating channel	2402 ~ 2480 MHz
RF transmit power range	-17.5 ~ 20 dBm

6.2.1 Bluetooth LE RF Transmitter (TX) Characteristics

Table 6-8. Bluetooth LE - Transmitter Characteristics - 1 Mbps

Parameter	Description	Min	Typ	Max	Unit
Carrier frequency offset and drift	Max. $ f_n _{n=0, 1, 2, 3, \dots k}$	—	0.5	—	kHz
	Max. $ f_0 - f_n _{n=2, 3, 4, \dots k}$	—	1.0	—	kHz
	Max. $ f_n - f_{n-5} _{n=6, 7, 8, \dots k}$	—	1.0	—	kHz
	$ f_1 - f_0 $	—	1.4	—	kHz
Modulation characteristics	$\Delta F1_{avg}$	—	256.3	—	kHz
	Min. $\Delta F2_{max}$ (for at least 99.9% of all $\Delta F2_{max}$)	—	252.6	—	kHz
	$\Delta F2_{avg}/\Delta F1_{avg}$	—	0.95	—	—
In-band emissions	± 2 MHz offset	—	-29	—	dBm
	± 3 MHz offset	—	-38	—	dBm
	$> \pm 3$ MHz offset	—	-44	—	dBm

Table 6-9. Bluetooth LE - Transmitter Characteristics - 2 Mbps

Parameter	Description	Min	Typ	Max	Unit
Carrier frequency offset and drift	Max. $ f_n _{n=0, 1, 2, 3, \dots k}$	—	0.5	—	kHz
	Max. $ f_0 - f_n _{n=2, 3, 4, \dots k}$	—	1.3	—	kHz
	Max. $ f_n - f_{n-5} _{n=6, 7, 8, \dots k}$	—	0.9	—	kHz
	$ f_1 - f_0 $	—	0.5	—	kHz
Modulation characteristics	$\Delta F1_{avg}$	—	508.2	—	kHz

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Table 6-9 – cont'd from previous page

Parameter	Description	Min	Typ	Max	Unit
	Min. $\Delta F_{2\max}$ (for at least 99.9% of all $\Delta F_{2\max}$)	—	516.9	—	kHz
	$\Delta F_{2\text{avg}}/\Delta F_{1\text{avg}}$	—	0.97	—	—
In-band emissions	± 4 MHz offset	—	-43	—	dBm
	± 5 MHz offset	—	-45	—	dBm
	$> \pm 5$ MHz offset	—	-48	—	dBm

Table 6-10. Bluetooth LE - Transmitter Characteristics - 125 Kbps

Parameter	Description	Min	Typ	Max	Unit
Carrier frequency offset and drift	Max. $ f_n _{n=0, 1, 2, 3, \dots k}$	—	1.3	—	kHz
	Max. $ f_0 - f_n _{n=1, 2, 3, \dots k}$	—	0.5	—	kHz
	$ f_0 - f_3 $	—	0.1	—	kHz
	Max. $ f_n - f_{n-3} _{n=7, 8, 9, \dots k}$	—	0.6	—	kHz
Modulation characteristics	$\Delta F_{1\text{avg}}$	—	255.7	—	kHz
	Min. $\Delta F_{1\max}$ (for at least 99.9% of all $\Delta F_{1\max}$)	—	261.9	—	kHz
In-band emissions	± 2 MHz offset	—	-26	—	dBm
	± 3 MHz offset	—	-36	—	dBm
	$> \pm 3$ MHz offset	—	-40	—	dBm

Table 6-11. Bluetooth LE - Transmitter Characteristics - 500 Kbps

Parameter	Description	Min	Typ	Max	Unit
Carrier frequency offset and drift	Max. $ f_n _{n=0, 1, 2, 3, \dots k}$	—	0.7	—	kHz
	Max. $ f_0 - f_n _{n=1, 2, 3, \dots k}$	—	0.5	—	kHz
	$ f_0 - f_3 $	—	0.1	—	kHz
	Max. $ f_n - f_{n-3} _{n=7, 8, 9, \dots k}$	—	0.6	—	kHz
Modulation characteristics	$\Delta F_{2\text{avg}}$	—	245.7	—	kHz
	Min. $\Delta F_{2\max}$ (for at least 99.9% of all $\Delta F_{2\max}$)	—	252.9	—	kHz
In-band emissions	± 2 MHz offset	—	-30	—	dBm
	± 3 MHz offset	—	-40	—	dBm
	$> \pm 3$ MHz offset	—	-45	—	dBm

6.2.2 Bluetooth LE RF Receiver (RX) Characteristics

Table 6-12. Bluetooth LE - Receiver Characteristics - 1 Mbps

Parameter	Description	Min	Typ	Max	Unit
Sensitivity @30.8% PER	—	—	-98.0	—	dBm
Maximum received signal @30.8% PER	—	—	8	—	dBm

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Table 6-12 – cont'd from previous page

Parameter		Description	Min	Typ	Max	Unit
C/I and receiver selectivity performance	Co-channel	$F = F_0$ MHz	—	7	—	dB
	Adjacent channel	$F = F_0 + 1$ MHz	—	–2	—	dB
		$F = F_0 - 1$ MHz	—	–4	—	dB
		$F = F_0 + 2$ MHz	—	–29	—	dB
		$F = F_0 - 2$ MHz	—	–32	—	dB
		$F = F_0 + 3$ MHz	—	–30	—	dB
		$F = F_0 - 3$ MHz	—	–43	—	dB
		$F \geq F_0 + 4$ MHz	—	–31	—	dB
		$F \leq F_0 - 4$ MHz	—	–46	—	dB
	Image frequency	—	—	–20	—	dB
Out-of-band blocking performance	Adjacent channel to image frequency	$F = F_{image} + 1$ MHz	—	–31	—	dB
		$F = F_{image} - 1$ MHz	—	–30	—	dB
		30 MHz ~ 2000 MHz	—	–25	—	dBm
		2003 MHz ~ 2399 MHz	—	–30	—	dBm
		2484 MHz ~ 2997 MHz	—	–30	—	dBm
		3000 MHz ~ 12.75 GHz	—	–25	—	dBm
Intermodulation		—	—	–43	—	dBm

Table 6-13. Bluetooth LE - Receiver Characteristics - 2 Mbps

Parameter		Description	Min	Typ	Max	Unit
Sensitivity @30.8% PER		—	—	–94.5	—	dBm
Maximum received signal @30.8% PER		—	—	8	—	dBm
C/I and receiver selectivity performance	Co-channel	$F = F_0$ MHz	—	9	—	dB
	Adjacent channel	$F = F_0 + 2$ MHz	—	–6	—	dB
		$F = F_0 - 2$ MHz	—	–8	—	dB
		$F = F_0 + 4$ MHz	—	–18	—	dB
		$F = F_0 - 4$ MHz	—	–36	—	dB
		$F = F_0 + 6$ MHz	—	–37	—	dB
		$F = F_0 - 6$ MHz	—	–42	—	dB
		$F \geq F_0 + 8$ MHz	—	–45	—	dB
		$F \leq F_0 - 8$ MHz	—	–44	—	dB
	Image frequency	—	—	–18	—	dB
Out-of-band blocking performance	Adjacent channel to image frequency	$F = F_{image} + 2$ MHz	—	–37	—	dB
		$F = F_{image} - 2$ MHz	—	–6	—	dB
		30 MHz ~ 2000 MHz	—	–25	—	dBm
		2003 MHz ~ 2399 MHz	—	–30	—	dBm
		2484 MHz ~ 2997 MHz	—	–30	—	dBm
		3000 MHz ~ 12.75 GHz	—	–25	—	dBm
Intermodulation		—	—	–31	—	dBm

Table 6-14. Bluetooth LE - Receiver Characteristics - 125 Kbps

Parameter		Description	Min	Typ	Max	Unit
Sensitivity @30.8% PER		—	—	−105.0	—	dBm
Maximum received signal @30.8% PER		—	—	8	—	dBm
C/I and receiver selectivity performance	Co-channel	$F = F_0 \text{ MHz}$	—	2	—	dB
	Adjacent channel	$F = F_0 + 1 \text{ MHz}$	—	−6	—	dB
		$F = F_0 - 1 \text{ MHz}$	—	−3	—	dB
		$F = F_0 + 2 \text{ MHz}$	—	−26	—	dB
		$F = F_0 - 2 \text{ MHz}$	—	−37	—	dB
		$F = F_0 + 3 \text{ MHz}$	—	−33	—	dB
		$F = F_0 - 3 \text{ MHz}$	—	−51	—	dB
		$F \geq F_0 + 4 \text{ MHz}$	—	−33	—	dB
		$F \leq F_0 - 4 \text{ MHz}$	—	−54	—	dB
	Image frequency	—	—	−28	—	dB
	Adjacent channel to image frequency	$F = F_{image} + 1 \text{ MHz}$	—	−33	—	dB
		$F = F_{image} - 1 \text{ MHz}$	—	−33	—	dB

Table 6-15. Bluetooth LE - Receiver Characteristics - 500 Kbps

Parameter		Description	Min	Typ	Max	Unit
Sensitivity @30.8% PER		—	—	−101.0	—	dBm
Maximum received signal @30.8% PER		—	—	8	—	dBm
C/I and receiver selectivity performance	Co-channel	$F = F_0 \text{ MHz}$	—	4	—	dB
	Adjacent channel	$F = F_0 + 1 \text{ MHz}$	—	−5	—	dB
		$F = F_0 - 1 \text{ MHz}$	—	−4	—	dB
		$F = F_0 + 2 \text{ MHz}$	—	−31	—	dB
		$F = F_0 - 2 \text{ MHz}$	—	−42	—	dB
		$F = F_0 + 3 \text{ MHz}$	—	−34	—	dB
		$F = F_0 - 3 \text{ MHz}$	—	−55	—	dB
		$F \geq F_0 + 4 \text{ MHz}$	—	−32	—	dB
		$F \leq F_0 - 4 \text{ MHz}$	—	−58	—	dB
	Image frequency	—	—	−24	—	dB
	Adjacent channel to image frequency	$F = F_{image} + 1 \text{ MHz}$	—	−32	—	dB
		$F = F_{image} - 1 \text{ MHz}$	—	−34	—	dB

6.3 Bluetooth Classic Radio

Table 6-16. Bluetooth Classic RF Characteristics

Name	Description
Center frequency range of operating channel	2402 ~ 2480 MHz
RF transmit power range	-17.5 ~ 12.5 dBm

6.3.1 Bluetooth Classic RF Transmitter (TX) Characteristics

Table 6-17. Bluetooth Classic - Transmitter Characteristics - Basic Rate (BR)

Parameter	Description	Min	Typ	Max	Unit
20 dB bandwidth	—	—	0.8	—	MHz
Adjacent channel transmit power	$F = F_0 \pm 2 \text{ MHz}$	—	-31.0	—	dBm
	$F = F_0 \pm 3 \text{ MHz}$	—	-44.0	—	dBm
	$F = F_0 \pm > 3 \text{ MHz}$	—	-48.0	—	dBm
Modulation characteristics	$\Delta F_{1\text{avg}}$	—	155.0	—	kHz
	Min. $\Delta F_{2\text{max}}$ (for at least 99.9% of all $\Delta F_{2\text{max}}$)	—	160.0	—	kHz
	$\Delta F_{2\text{avg}}/\Delta F_{1\text{avg}}$	—	0.96	—	—
ICFT	—	—	1.0	—	kHz
Drift rate	—	—	0.5	—	kHz/50 μs
Drift (DH1)	—	—	0.5	—	kHz
Drift (DH5)	—	—	0.5	—	kHz

Table 6-18. Bluetooth Classic - Transmitter Characteristics - Enhanced Data Rate (EDR)

Parameter	Description	Min	Typ	Max	Unit
$\pi/4$ DQPSK max w_0	—	—	-2.0	—	kHz
$\pi/4$ DQPSK max w_i	—	—	-15.0	—	kHz
$\pi/4$ DQPSK max $ w_i + w_0 $	—	—	-17.0	—	kHz
8DPSK max w_0	—	—	-4.0	—	kHz
8DPSK max w_i	—	—	-17.0	—	kHz
8DPSK max $ w_i + w_0 $	—	—	-21.0	—	kHz
$\pi/4$ DQPSK modulation accuracy	RMS DEVM	—	7.0	—	%
	99% DEVM	—	0.1	—	%
	Peak DEVM	—	11.0	—	%
8DPSK modulation accuracy	RMS DEVM	—	7.0	—	%
	99% DEVM	—	0.1	—	%
	Peak DEVM	—	12.0	—	%
In-band emissions	$F = F_0 \pm 1 \text{ MHz}$	—	-35	—	dBm
	$F = F_0 \pm 2 \text{ MHz}$	—	-25	—	dBm

Cont'd on next page

Table 6-18 – cont'd from previous page

Parameter	Description	Min	Typ	Max	Unit
	$F = F_0 \pm 3 \text{ MHz}$	—	–39	—	dBm
	$F = F_0 \pm > 3 \text{ MHz}$	—	–43	—	dBm

6.3.2 Bluetooth Classic RF Receiver (RX) Characteristics

Table 6-19. Bluetooth Classic - Receiver Characteristics - Basic Rate (BR)

Parameter	Description	Min	Typ	Max	Unit
Sensitivity with dirty transmit off @0.1% BER	—	—	–95.0	—	dBm
Maximum received signal @0.1% BER	—	—	8	—	dBm
C/I and receiver selectivity performance	Co-channel	$F = F_0 \text{ MHz}$	—	8	dB
	Adjacent channel	$F = F_0 + 1 \text{ MHz}$	—	–7	dB
		$F = F_0 - 1 \text{ MHz}$	—	–8	dB
		$F = F_0 + 2 \text{ MHz}$	—	–30	dB
		$F = F_0 - 2 \text{ MHz}$	—	–27	dB
		$F = F_0 + 3 \text{ MHz}$	—	–36	dB
		$F = F_0 - 3 \text{ MHz}$	—	–40	dB
	Image frequency	—	—	–29	dB
	Adjacent channel to image frequency	$F = F_{\text{image}} + 1 \text{ MHz}$	—	–40	dB
		$F = F_{\text{image}} - 1 \text{ MHz}$	—	–36	dB
Out-of-band blocking performance	30 MHz ~ 2000 MHz	—	–5	—	dBm
	2003 MHz ~ 2399 MHz	—	–22	—	dBm
	2484 MHz ~ 2997 MHz	—	–22	—	dBm
	3000 MHz ~ 12.75 GHz	—	–5	—	dBm
Intermodulation	—	—	–37	—	dBm

Table 6-20. Bluetooth Classic - Receiver Characteristics - Enhanced Data Rate (EDR)

Parameter	Description	Min	Typ	Max	Unit
$\pi/4$ DQPSK					
Sensitivity with dirty transmit off @0.01% BER	—	—	–94.0	—	dBm
Maximum received signal @0.01% BER	—	—	8	—	dBm
C/I and receiver selectivity performance	Co-channel	$F = F_0 \text{ MHz}$	—	10	dB
	Adjacent channel	$F = F_0 + 1 \text{ MHz}$	—	–10	dB
		$F = F_0 - 1 \text{ MHz}$	—	–11	dB
		$F = F_0 + 2 \text{ MHz}$	—	–36	dB
		$F = F_0 - 2 \text{ MHz}$	—	–28	dB
		$F = F_0 + 3 \text{ MHz}$	—	–35	dB
		$F = F_0 - 3 \text{ MHz}$	—	–40	dB
	Image frequency	—	—	–27	dB
	Adjacent channel to image frequency	$F = F_{\text{image}} + 1 \text{ MHz}$	—	–40	dB
		$F = F_{\text{image}} - 1 \text{ MHz}$	—	–35	dB

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Table 6-20 – cont'd from previous page

Parameter		Description	Min	Typ	Max	Unit
8DPSK						
Sensitivity with dirty transmit off @0.01% BER		—	—	−87.0	—	dBm
Maximum received signal @0.01% BER		—	—	2	—	dBm
C/I and receiver selectivity performance	Co-channel	$F = F_0$ MHz	—	18	—	dB
	Adjacent channel	$F = F_0 + 1$ MHz	—	−3	—	dB
		$F = F_0 - 1$ MHz	—	−4	—	dB
		$F = F_0 + 2$ MHz	—	−28	—	dB
		$F = F_0 - 2$ MHz	—	−21	—	dB
		$F = F_0 + 3$ MHz	—	−33	—	dB
		$F = F_0 - 3$ MHz	—	−37	—	dB
	Image frequency	—	—	−20	—	dB
	Adjacent channel to image frequency	$F = F_{\text{image}} + 1$ MHz	—	−33	—	dB
		$F = F_{\text{image}} - 1$ MHz	—	−33	—	dB

6.4 802.15.4 Radio

Table 6-21. 802.15.4 RF Characteristics

Name	Description
Center frequency range of operating channel	2405 ~ 2480 MHz

¹ Zigbee in the 2.4 GHz range supports 16 channels at 5 MHz spacing from channel 11 to channel 26.

6.4.1 802.15.4 RF Transmitter (TX) Characteristics

Table 6-22. 802.15.4 Transmitter Characteristics - 250 Kbps

Parameter	Min	Typ	Max	Unit
RF transmit power range	−17.5	—	19.0	dBm
EVM	—	24.0%	—	—

6.4.2 802.15.4 RF Receiver (RX) Characteristics

Table 6-23. 802.15.4 Receiver Characteristics - 250 Kbps

Parameter	Description	Min	Typ	Max	Unit
Sensitivity @1% PER	—	—	−102.0	—	dBm
Maximum received signal @1% PER	—	—	8	—	dBm
Relative jamming level	Adjacent channel	$F = F_0 + 5$ MHz	—	−30	—
				dB	

Cont'd on next page

Table 6-23 – cont'd from previous page

Parameter		Description	Min	Typ	Max	Unit
		$F = F_0 - 5 \text{ MHz}$	—	–40	—	dB
	Alternate channel	$F = F_0 + 10 \text{ MHz}$	—	–52	—	dB
		$F = F_0 - 10 \text{ MHz}$	—	–53	—	dB

7 Packaging

- For information about tape, reel, and chip marking, please refer to [ESP32-S31 Chip Packaging Information](#).
- The pins of the chip are numbered in anti-clockwise order starting from Pin 1 in the top view. For pin numbers and pin names, see also Figure 2-1 *ESP32-S31 Pin Layout (Top View)*.

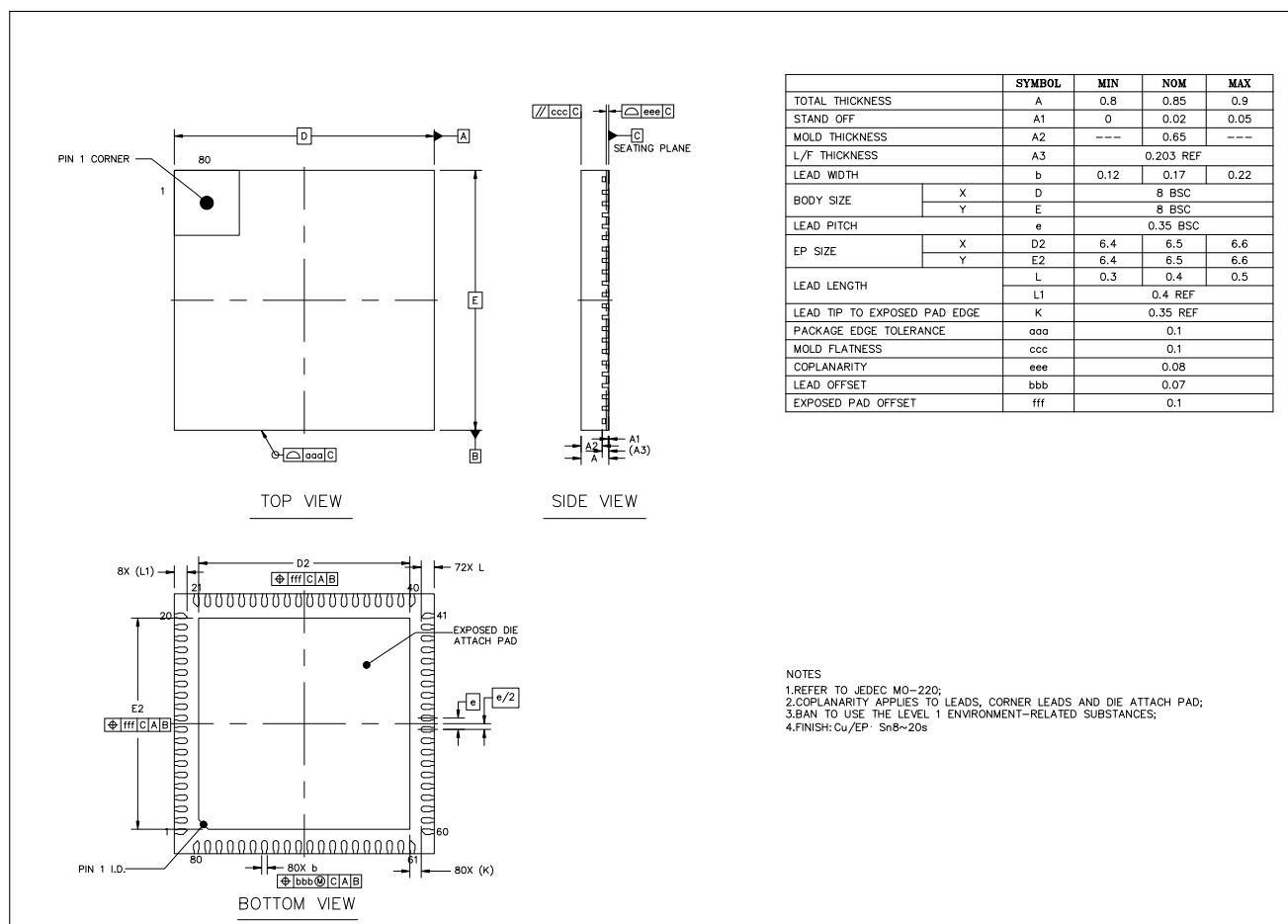


Figure 7-1. QFN80 (8×8 mm) Package

ESP32-S31 Consolidated Pin Overview

Table 7-1. Consolidated Pin Overview

Pin No.	Pin Name	Pin Type	Pin Providing Power	Pin Settings		Analog Functions	F0	Type	F1	LP IO MUX Functions		Type	F3	Type	IO MUX Functions									
				At Reset	After Reset					F2	F0				Type	F1	Type	F2	Type	F3	Type	F4	Type	
1	ANT	Analog																						
2	VDDA3	Power																						
3	VDDA4	Power																						
4	CHIP_PU	Analog																						
5	XTAL_32K_N/GPIO0	IO	VDDPST_1			XTAL_32K_N	LP_GPIO0	I/O/T	LP_GPIO0	I/O/T			LP_PROBE_TOP_OUT0	O	GPIO0	I/O/T	GPIO0	I/O/T						
6	XTAL_32K_P/GPIO1	IO	VDDPST_1			XTAL_32K_P	LP_GPIO1	I/O/T	LP_GPIO1	I/O/T			LP_PROBE_TOP_OUT1	O	GPIO1	I/O/T	GPIO1	I/O/T						
7	GPIO2	IO	VDDPST_1				LP_UART_DTRN_PAD	O	LP_GPIO2	I/O/T	LP_SPI_CK_PAD	I/O/T	LP_PROBE_TOP_OUT2	O	GPIO2	I/O/T	GPIO2	I/O/T			LCD_DATA19_OUT_PAD	O		
8	GPIO3	IO	VDDPST_1				LP_UART_DSRN_PAD	I	LP_GPIO3	I/O/T	LP_SPI_CS_PAD	I/O/T	LP_PROBE_TOP_OUT3	O	GPIO3	I/O/T	GPIO3	I/O/T			LCD_DATA20_OUT_PAD	O		
9	GPIO4	IO	VDDPST_1				LP_UART_RTSN_PAD	O	LP_GPIO4	I/O/T	LP_SPI_D_PAD	I/O/T	LP_PROBE_TOP_OUT4	O	GPIO4	I/O/T	GPIO4	I/O/T			LCD_DATA21_OUT_PAD	O		
10	GPIO5	IO	VDDPST_1				LP_UART_CTSN_PAD	I	LP_GPIO5	I/O/T	LP_SPI_Q_PAD	I/O/T	LP_PROBE_TOP_OUT5	O	GPIO5	I/O/T	GPIO5	I/O/T			LCD_DATA22_OUT_PAD	O		
11	VDDPST_1	Power																						
12	GPIO6	IO	VDDPST_1			TOUCH_CH0	LP_UART_TXD_PAD	O	LP_GPIO6	I/O/T	LP_I2C_SCL_PAD	I/O/T	LP_PROBE_TOP_OUT6	O	GPIO6	I/O/T	GPIO6	I/O/T						
13	GPIO7	IO	VDDPST_1			TOUCH_CH1	LP_UART_RXD_PAD	I	LP_GPIO7	I/O/T	LP_I2C_SDA_PAD	I/O/T	LP_PROBE_TOP_OUT7	O	GPIO7	I/O/T	GPIO7	I/O/T			LCD_DATA23_OUT_PAD	O		
14	GPIO8	IO	VDDPST_1		IE	TOUCH_CH2									GPIO8	I/O/T	GPIO8	I/O/T	GMAC_PHY_TXD0_PAD	O	LCD_DATA24_OUT_PAD	O		
15	GPIO9	IO	VDDPST_1		IE	TOUCH_CH3									SPI2_HOLD_PAD	I/O/T	GPIO9	I/O/T	GMAC_PHY_TXD1_PAD	O	LCD_DATA25_OUT_PAD	O		
16	GPIO10	IO	VDDPST_1		IE	TOUCH_CH4									SPI2_CS_PAD	I/O/T	GPIO10	I/O/T	GMAC_PHY_TXD2_PAD	O	LCD_DATA26_OUT_PAD	O		
17	GPIO11	IO	VDDPST_1		IE	TOUCH_CH5									SPI2_D_PAD	I/O/T	GPIO11	I/O/T	GMAC_PHY_TXD3_PAD	O	LCD_DATA27_OUT_PAD	O		
18	VREF_TOUCH	Analog																						
19	GPIO12	IO	VDDPST_1		IE	TOUCH_CH6									SPI2_CK_PAD	I/O/T	GPIO12	I/O/T	GMAC_PHY_TXEN_PAD	O	LCD_DATA28_OUT_PAD	O		
20	GPIO13	IO	VDDPST_1		IE	TOUCH_CH7									SPI2_Q_PAD	I/O/T	GPIO13	I/O/T	GMAC_RMII_CLK_PAD	I/O/T	LCD_DATA29_OUT_PAD	O		
21	GPIO14	IO	VDDPST_1		IE	TOUCH_CH8									SPI2_WP_PAD	I/O/T	GPIO14	I/O/T	GMAC_RX_CLK_PAD	IO	LCD_DATA30_OUT_PAD	O		
22	GPIO15	IO	VDDPST_1		IE	TOUCH_CH9									SPI2_I04_PAD	I/O/T	GPIO15	I/O/T	GMAC_PHY_RXD0_PAD	IO	LCD_DATA31_OUT_PAD	O		
23	GPIO16	IO	VDDPST_1		IE	TOUCH_CH10									SPI2_I05_PAD	I/O/T	GPIO16	I/O/T	GMAC_PHY_RXD3_PAD	IO	LCD_DATA32_OUT_PAD	O		
24	GPIO17	IO	VDDPST_1		IE	TOUCH_CH11									SPI2_I06_PAD	I/O/T	GPIO17	I/O/T	GMAC_PHY_RXD2_PAD	IO	LCD_DATA33_OUT_PAD	O		
25	GPIO18	IO	VDDPST_1		IE	TOUCH_CH12									SPI2_I07_PAD	I/O/T	GPIO18	I/O/T	GMAC_PHY_RXD1_PAD	IO	LCD_DATA34_OUT_PAD	O		
26	GPIO19	IO	VDDPST_1		IE	TOUCH_CH13									SPI2_DQS_PAD	O/T	GPIO19	I/O/T	GMAC_PHY_RXD0_PAD	IO	LCD_DATA35_OUT_PAD	O		
27	SDIO_DATA0	IO	VDDPST_SD		IE										SDIO_DATA0	I/O/T	GPIO20	I/O/T	SPI2_CK_PAD	I/O/T				
28	SDIO_DATA1	IO	VDDPST_SD		IE										SDIO_DATA1	I/O/T	GPIO21	I/O/T	SPI2_D_PAD	I/O/T				
29	SDIO_DATA2	IO	VDDPST_SD		IE										SDIO_DATA2	I/O/T	GPIO22	I/O/T	SPI2_Q_PAD	I/O/T				
30	VDD_PSRAM_1P8_1	Power																						
31	SDIO_DATA3	IO	VDDPST_SD		IE										SDIO_DATA3	I/O/T	GPIO23	I/O/T	SPI2_CS_PAD	I/O/T				
32	SDIO_CLK	IO	VDDPST_SD		IE										SDIO_CLK	O	GPIO24	I/O/T	SPI2_HOLD_PAD	I/O/T				
33	SDIO_CMD	IO	VDDPST_SD		IE										SDIO_CMD	I/O/T	GPIO25	I/O/T	SPI2_WP_PAD	I/O/T				
34	VDD_PSRAM_1P8_2	Power																						
35	VDD_LDO_1P8	Power																						
36	SPICS	IO	VDD_SPI		WPU	WPU, IE									SPICS	O/T	GPIO26	I/O/T						
37	SPIQ	IO	VDD_SPI		WPU	WPU, IE									SPIQ	I/O/T	GPIO27	I/O/T						
38	SPIWP	IO	VDD_SPI		WPU	WPU, IE									SPIWP	I/O/T	GPIO28	I/O/T						
39	VDD_SPI	Power																						
40	SPIHD	IO	VDD_SPI		WPU	WPU, IE									SPIHD	I/O/T	GPIO30	I/O/T						
41	SPICLK	IO	VDD_SPI		WPU	WPU, IE									SPICLK	O/T	GPIO31	I/O/T						
42	SPID	IO	VDD_SPI		WPU	WPU, IE									SPID	I/O/T	GPIO32	I/O/T						
43	VCCA/VDDPST_2	Power																						
44	USB_DP	Analog	VDDPST_2																					
45	USB_DM	Analog	VDDPST_2																					
46	GPIO33	IO	VDDPST_3	DRV=3	USB_FU, IE, DRV=3	USB1P1_NO									GPIO33	I/O/T	GPIO33	I/O/T			LCD_DATA36_OUT_PAD	O		
47	GPIO34	IO	VDDPST_3	DRV=3	USB_FU, IE, DRV=3	USB1P1_PO									GPIO34	I/O/T	GPIO34	I/O/T			LCD_DATA37_OUT_PAD	O		
48	GPIO35	IO	VDDPST_3		IE										GPIO35	I/O/T	GPIO35	I/O/T	REF_GMAC_CLK_PAD	O	LCD_DATA38_OUT_PAD	O	SD2_CDATA0_PAD	I/O/T
49	GPIO36	IO	VDDPST_3		IE										GPIO36	I/O/T	GPIO36	I/O/T			LCD_DATA39_OUT_PAD	O	SD2_CDATA1_PAD	I/O/T
50	GPIO37	IO	VDDPST_3		IE	PAD_COMP0									GPIO37	I/O/T	GPIO37	I/O/T			LCD_DATA40_OUT_PAD	O	SD2_CDATA2_PAD	I/O/T
51	GPIO38	IO	VDDPST_3		IE	PAD_COMP1									GPIO38	I/O/T	GPIO38	I/O/T			LCD_DATA41_OUT_PAD	O	SD2_CDATA3_PAD	I/O/T
52	GPIO39	IO	VDDPST_3		IE	PAD_COMP2									GPIO39	I/O/T	GPIO39	I/O/T			LCD_DATA42_OUT_PAD	O	SD2_CCLK_PAD	O
53	GPIO40	IO	VDDPST_3		IE	PAD_COMP3									GPIO40	I/O/T	GPIO40	I/O/T			LCD_PCLK_PAD	O	SD2_CCMD_PAD	I/O/T
54	VDDPST_3	Power																						
55	GPIO42	IO	VDDPST_3		IE	ADC1_CH0_N									GPIO42	I/O/T	GPIO42	I/O/T						
56	GPIO43	IO	VDDPST_3		IE	ADC1_CH0_P									GPIO43	I/O/T	GPIO43	I/O/T			LCD_H_ENABLE_PAD	O		
57	GPIO44	IO	VDDPST_3		IE	ADC1_CH1_N									GPIO44	I/O/T	GPIO44	I/O/T			LCD_H_SYNC_PAD	O		
58	GPIO45	IO	VDDPST_3		IE	ADC1_CH1_P									GPIO45	I/O/T	GPIO45	I/O/T			LCD_V_SYNC_PAD	O		
59	GPIO46	IO	VDDPST_3		IE	ADC1_CH2_N									GPIO46	I/O/T	GPIO46	I/O/T			CAM_DATA0_IN_PAD	IO		
60	GPIO47	IO	VDDPST_3		IE	ADC1_CH2_P									GPIO47	I/O/T	GPIO47	I/O/T			CAM_DATA1_IN_PAD	IO		
61	GPIO48	IO	VDDPST_3			ADC1_CH3_N									GPIO48	I/O/T	GPIO48	I/O/T			CAM_DATA2_IN_PAD	IO		
62	GPIO49	IO	VDDPST_3			ADC1_CH3_P									GPIO49	I/O/T	GPIO49	I/O/T			CAM_DATA3_IN_PAD	IO		
63	VREF_ADC	Analog																						
64	VDDPST_4	Power																						
65	GPIO50	IO	VDDPST_4			ADC2_CH0_N									GPIO50	I/O/T	GPIO50	I/O/T			CAM_DATA4_IN_PAD	IO		
66	GPIO51	IO	VDDPST_4			ADC2_CH0_P									GPIO51	I/O/T	GPIO51	I/O/T			CAM_DATA5_IN_PAD	IO		
67	GPIO52	IO	VDDPST_4		IE	ADC2_CH1_N									GPIO52	I/O/T	GPIO52	I/O/T	SPI2_CS_PAD	I/O/T	CAM_DATA6_IN_PAD	IO		
68	GPIO53	IO	VDDPST_4		IE	ADC2_CH1_P									GPIO53	I/O/T	GPIO53	I/O/T	SPI2_CK_PAD	I/O/T	CAM_DATA7_IN_PAD	IO		
69	MTD0	IO	VDDPST_4		IE	ADC2_CH2_N									MTD0	O/T	GPIO54	I/O/T	SPI2_D_PAD	I/O/T	CAM_PCLK_PAD	IO		
70	MTCK	IO	VDDPST_4			ADC2_CH2_P									MTCK	I	GPIO55	I/O/T	SPI2_Q_PAD	I/O/T	CAM_XCLK_PAD	O		
71	MTDI	IO	VDDPST_4		IE	ADC2_CH3_N									MTDI	I	GPIO56	I/O/T	SPI2_HOLD_PAD	I/O/T	CAM_V_SYNC_PAD	IO		
72	MTMS	IO	VDDPST_4		IE	ADC2_CH3_P									MTMS	I	GPIO57	I/O/T	SPI2_WP_PAD	I/O/T	CAM_H_SYNC_PAD	IO		
73	GPIO58	IO	VDDPST_4		IE										UART0_TXD_PAD	O	GPIO58	I/O/T						

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Pin No.	Pin Name	Pin Type	Pin Providing Power	Pin Settings		Analog Functions	LP IO MUX Functions								IO MUX Functions													
				At Reset	After Reset		F0	Type	F1	Type	F2	Type	F3	Type	F0	Type	F1	Type	F2	Type	F3	Type	F4	Type				
74	GPIO59	IO	VDDPST_4		IE										UART0_RXD_PAD	I1	GPIO59	I/O/T										
75	GPIO60	IO	VDDPST_4		WPU, IE										GPIO60	I/O/T	GPIO60	I/O/T										
76	GPIO61	IO	VDDPST_4		WPU, IE										GPIO61	I/O/T	GPIO61	I/O/T										
77	VDDA1	Power																										
78	XTAL_N	Analog																										
79	XTAL_P	Analog																										
80	VDDA2	Power																										
81	GND	Power																										

* For details, see Section 2 *Pins*. Regarding highlighted cells, see Section 2.3.4 *Restrictions for GPIOs and LP GPIOs*.

Glossary

strapping pin

A type of GPIO pin used to configure certain operational settings during the chip's power-up, and can be reconfigured as normal GPIO after the chip's reset [36](#)

eFuse parameter

A parameter stored in an electrically programmable fuse (eFuse) memory within a chip. The parameter can be set by programming EFUSE_PGM_DATA_n_REG registers, and read by reading a register field named after the parameter [36](#)

SPI boot mode

A boot mode in which users load and execute the existing code from SPI flash [37](#)

joint download boot mode

A boot mode in which users can download code into flash via the UART or other interfaces (see Table [3-3 Chip Boot Mode Control](#) > Note), and load and execute the downloaded code from the flash or SRAM [37](#), [38](#)

Related Documentation and Resources

Related Documentation

- [ESP32-S31 Hardware Design Guidelines](#) – Guidelines on how to integrate the ESP32-S31 into your hardware product.
- [ESP32-S31 Series SoC Errata](#) – Descriptions of known errors in ESP32-S31 series of SoCs.
- [Certificates](#)
<https://espressif.com/en/support/documents/certificates>
- [ESP32-S31 Product/Process Change Notifications \(PCN\)](#)
<https://espressif.com/en/support/documents/pcns?keys=ESP32-s31>
- [ESP32-S31 Advisories](#) – Information on security, bugs, compatibility, component reliability.
<https://espressif.com/en/support/documents/advisories?keys=ESP32-s31>
- [Documentation Updates and Update Notification Subscription](#)
<https://espressif.com/en/support/download/documents>

Developer Zone

- [ESP-IDF Programming Guide for ESP32-S31](#) – Extensive documentation for the ESP-IDF development framework.
- [ESP-IDF](#) and other development frameworks on GitHub.
<https://github.com/espressif>
- [ESP32 BBS Forum](#) – Engineer-to-Engineer (E2E) Community for Espressif products where you can post questions, share knowledge, explore ideas, and help solve problems with fellow engineers.
<https://esp32.com/>
- [ESP-FAQ](#) – A summary document of frequently asked questions released by Espressif.
<https://espressif.com/projects/esp-faq/en/latest/index.html>
- [The ESP Journal](#) – Best Practices, Articles, and Notes from Espressif folks.
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Products

- [ESP32-S31 Series SoCs](#) – Browse through all ESP32-S31 SoCs.
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<https://products.espressif.com/#/product-selector?language=en>

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Revision History

Date	Version	Release notes
2026-07-13	v0.5	• Product Overview: Updated Functional Block Diagram and clarified processor architecture and feature descriptions to reflect HP dual-core and HP/LP dual-system architecture • 2 Pins: Updated pin-related tables and added 2-8 Peripheral Pin Assignment • 2.3.2 LP IO MUX Functions: Added table 2-4 LP Peripheral Signals Routed via LP IO MUX • 2.5.2 Power Scheme: Updated figure 2-2 ESP32-S31 Power Scheme • 2.6 Pin Mapping Between Chip and Flash: Corrected pin number errors in 2-13 Pin Mapping Between Chip and Off-Package Flash • 5.3 VDD_SPI Output Characteristics: Added table 5-3 VDD_SPI Internal and Output Characteristics • 5.5 Current Consumption: Added 5-9 Current Consumption in Modem-sleep Mode and 5-10 Current Consumption in Low-Power Modes • 6 RF Characteristics: Updated RF characteristics
2026-05-21	v0.2	• Removed "Confidential" classification for the ESP32-S31 chip • 3 Boot Configurations: Added 3.1.1 Secure Debugging in Joint Download Boot Mode • 4.2.2.9 CAN FD Controller: Updated from TWAI to CAN FD controller description • 5.5.1 RF Current Consumption in Active Mode: Added current consumption data in Active mode • 7 Packaging: Added QFN80 package diagram
2026-04-03	v0.1	Draft



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